

SERVICE MANUAL

**1541/1541-II
DISK DRIVE**

FEBRUARY, 1989

PN-314862-01



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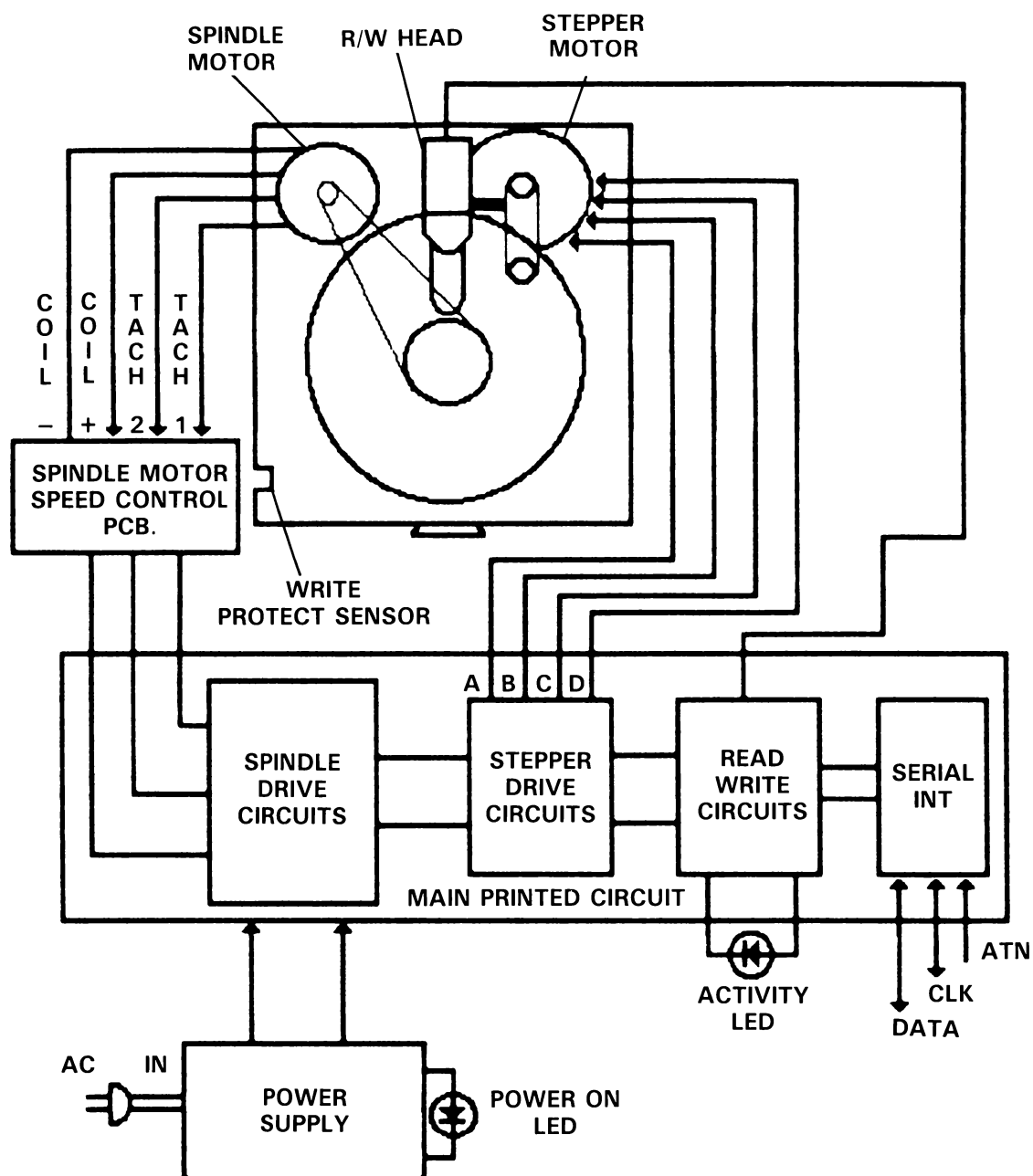
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C1541 DISK DRIVE PRODUCT SPECIFICATION

GENERAL DESCRIPTION	The C1541 Disk Drive is an external 5-1/4 inch floppy diskette recorder, offering high-speed and capacity for programs and data. It is an intelligent device, containing its own microprocessor, RAM, ROM and operating systems software for faster speed of throughput and memory efficiency in the computer.
MAXIMUM STORAGE	170K of data (formatted) — 35 tracks
MEDIA	5-1/4 Inch floppy disk. Single sided, single density, soft sectored (double density can be used, but not needed).
INPUT/OUTPUT	Commodore serial interface Second serial port for chaining a second drive or printer
CONTROLLER	MOS 6502 microprocessor — 1 MHz clock
MEMORY	2K RAM, 16K ROM
DATA TRANSFER RATE	400 Bytes/sec
FILE TYPES	Program, sequential, relative, random-access and user
NUMBER OF FILES	Up to 144 different files per diskette
COMPUTERS	C64, VIC 20, SX64, Educator 64, Plus/4, C16
MEDIA COMPATIBILITY	2031, 4040, C1551, C1571
POWER REQUIREMENTS	120 Volts AC, 60Hz — integral power supply with external 1 Amp fuse
POWER CONSUMPTION	30 Watts maximum

BLOCK DIAGRAM



CARE AND MAINTENANCE

- DO NOT use MAGNETIZED tools when repairing or adjusting a disk drive.
- DO NOT place a disk drive near any device which generates “noise” e.g., — motors, radios, televisions.
- DO NOT stack drives upon each other or in any way inhibit air flow around the unit. HEAT BUILD-UP can cause disk failures.
- Periodically CLEAN the read/write head with 90% isopropyl alcohol and a cotton swab. CHECK load pad for excess wear. Clean or replace as necessary.
- Take the following precautions when handling a diskette:
 - ALWAYS store a diskette in its jacket.
 - Use ONLY felt-tip pens when writing on the label of a diskette.
 - Do not bend or physically damage a diskette.
 - Do not place a diskette in the area of a magnetic field.
 - Do not attempt to clean a diskette.
 - Do not touch the exposed area of a diskette.
- DIAGNOSTIC and ADJUSTMENT procedures are outlined in detail on the diagnostic disk (Commodore Part #31405101). A manual has been added to the diagnostic package. It contains descriptions of testing procedures and adjustment methods.

OVERVIEW

The drive is itself an independent memory device. The drive is composed of a media clamp rotating mechanism, a head positioning mechanism and an eject mechanism. All positioning operations, excluding insertion and removal of the diskette, are controlled by the internal guide mechanism. Closing the front door causes the media clamp mechanism to operate. Two operations are performed in the following order:

- a) The diskette is centered.
- b) The diskette is clamped and retained between the spindle and the hub.

The spindle and hub rotate at 300 r.p.m. through a closed-loop control circuit employing a D.C. motor/tachometer. It is important that the relationship between the head and the media is maintained correctly during operation. For this purpose, a pressure pad is used to hold and press down the media (about 12g) from the opposite side of the head. This head assembly is coupled by a metal band to a four phase stepping motor which performs the track positioning. One step of the stepping motor corresponds to a 1/2 track movement. The control circuit on the logic board selects the direction and number of steps to the desired track.

The Read/Write head uses a glass-bonded, ferrite/ceramic head. Track-to-track erasing is accomplished by the straddle erase method. The surface of the Read/Write head is mirror-ground to minimize wear of the head and media. Also, the head is designed in such a way that the maximum signal can be obtained from the media surface.

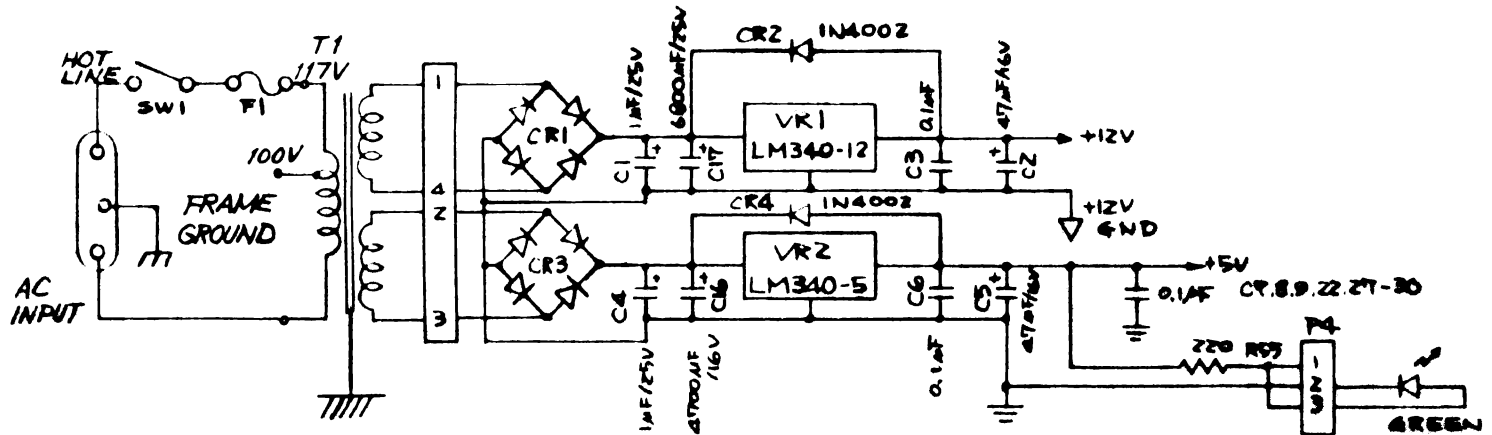
The spindle drive motor operates on 12 VDC and turns the spindle, through a belt drive, at 300 revolutions per minute. The speed of the drive motor is controlled by a feedback signal from a tachometer, which is housed in the drive motor assembly. The feedback signal controls a servo amp that supplies the 12 VDC drive current.

FLASH CODE

The 1541, upon power-up, goes through its own internal diagnostic. If an electronic problem is detected, it is indicated by a FLASH CODE. The LED's will blink a set number of times, pause, and then flash again until the problem is corrected.

Number of Flashes	Possible Failure
2	Zero Page
3,4	DOS ROM's
5,6,7,8	RAM

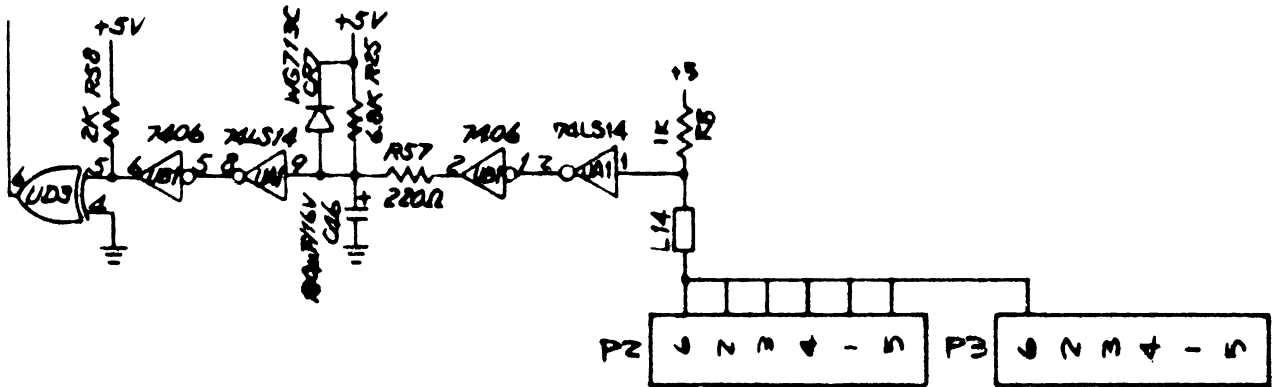
Circuitry associated with these components can also cause the failure code. Therefore, it should be suspected as the next possible defect.



All circuit diagrams have been taken from the short board schematic 1540049 unless otherwise noted. The short boards use a 6116 RAM which replaces the four 2114 I.C.s on the long board. See page 11 for the Read/Write logic differences.

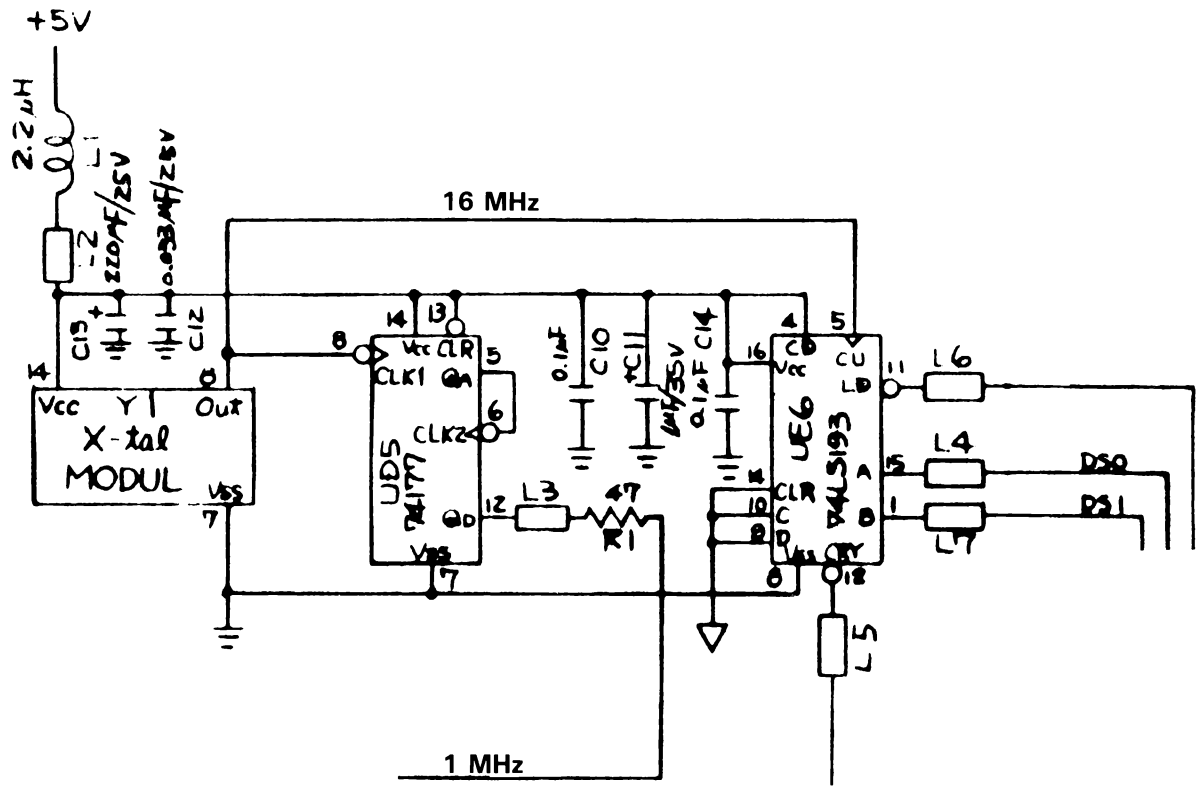
The Power Supply

The input AC voltage is controlled by switch 1 (SW1). Disk circuit protection is provided by fuse 1 (F1). If SW1 is closed, the AC voltage input is applied to the primary winding of transformer one (T1). T1 steps down the AC input voltage into two smaller AC voltages. The top secondary AC output (approx. 16VRMS) is converted to DC by the Full Wave Bridge Rectifier CR1. The DC output of CR1 is regulated at 12VDC by VR1. The bottom secondary AC output of T1 (approx. 9VRMS) is converted to DC by the Full Wave Bridge Rectifier CR3. The DC output of CR3 is regulated at +5VDC by VR2. High frequency filtering is provided by C1 and C3 for the 12VDC supply, and C4, C6 to C9, C22, C27 to C30 for the 5VDC supply. Low frequency filtering is provided by C17 and C2 for the 12VDC supply, and C5 and C16 for the 5VDC supply.



The Reset Circuit

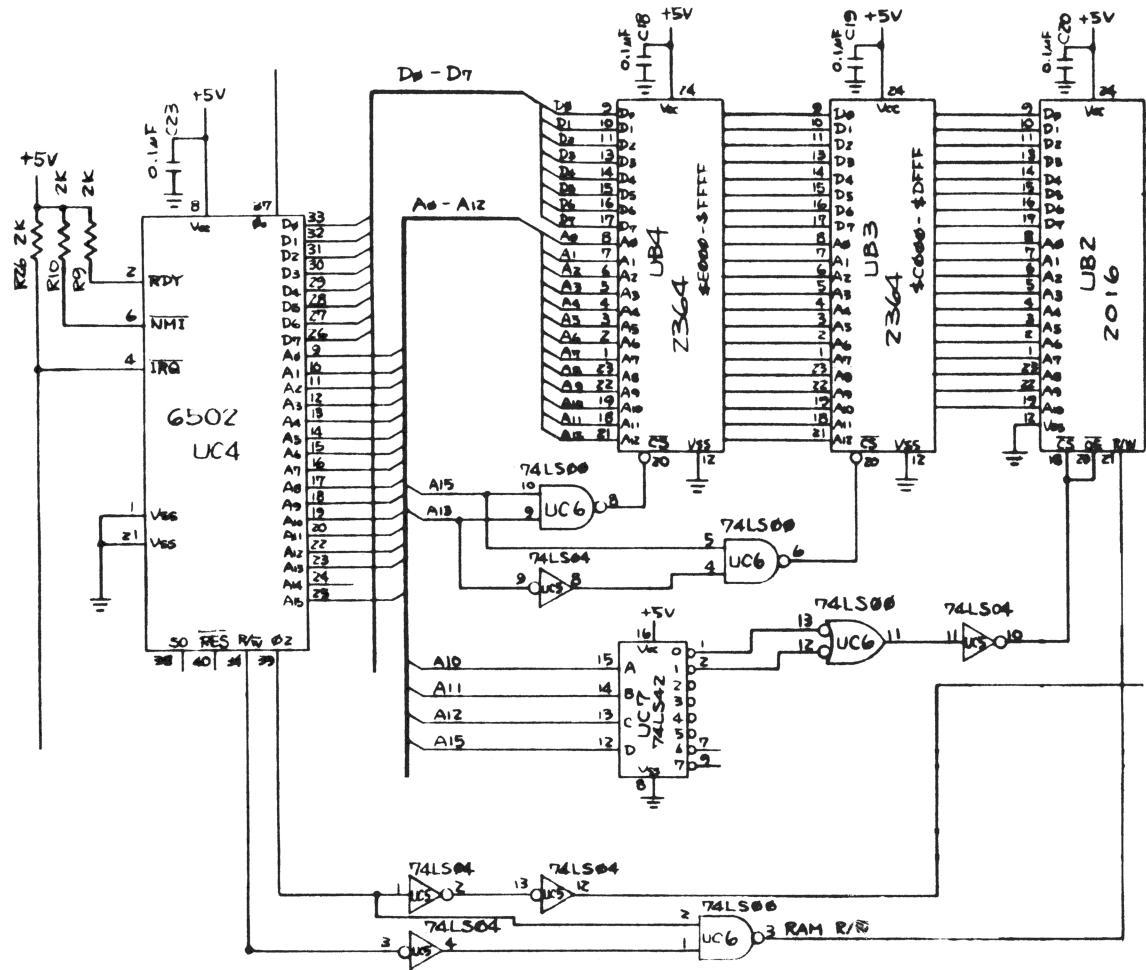
The output of the exclusive 'or' gate UD3 pin 6 will be "low" until C46 has charged through R25. Once the voltage across C46 reaches 2 volts, the output of UD3 pin 6 will go "high". This occurs when the disk is powered on, or a reset pulse is generated by a device connected to the serial bus. The reset pulse on the serial bus interface is input on, pin 6 of P2 or P3. This "low" to "high" going pulse on pin 6 of UD3 is input to the microprocessors reset interrupt input. This causes a restart or reset routine to be executed giving control of the disk drive operation to the Disk Operating System (DOS).



The Clock Circuits

Crystal Y1 outputs a 16 MHz clock signal. This is input to UD5 on pin 8. UD5 is configured as a $\div 16$ frequency divider. The output of UD5 pin 12 is a 1 MHz clock signal used as the system clock (Phase 0) for the microprocessor. UE6 is a programmable counter ($\div 16$, $\div 15$, $\div 14$, $\div 13$) that outputs a varying frequency clock used to compensate for the difference in recording area/sector for sectors on inner tracks (Trks 1,2,3) as compared to sectors on out most tracks (Trks 33,34,35). The area/sector for inner tracks is less than the area/sector for out most tracks, so the recording clock frequency is increased when writing on inner tracks to keep the flux density constant. This clock output is on pin 12 of UE6.

Tracks	Clock Frequency	Divide By
1-17	1.2307 MHz	13
18-24	1.1428 MHz	14
25-30	1.0666 MHz	15
31-35	1 MHz	16

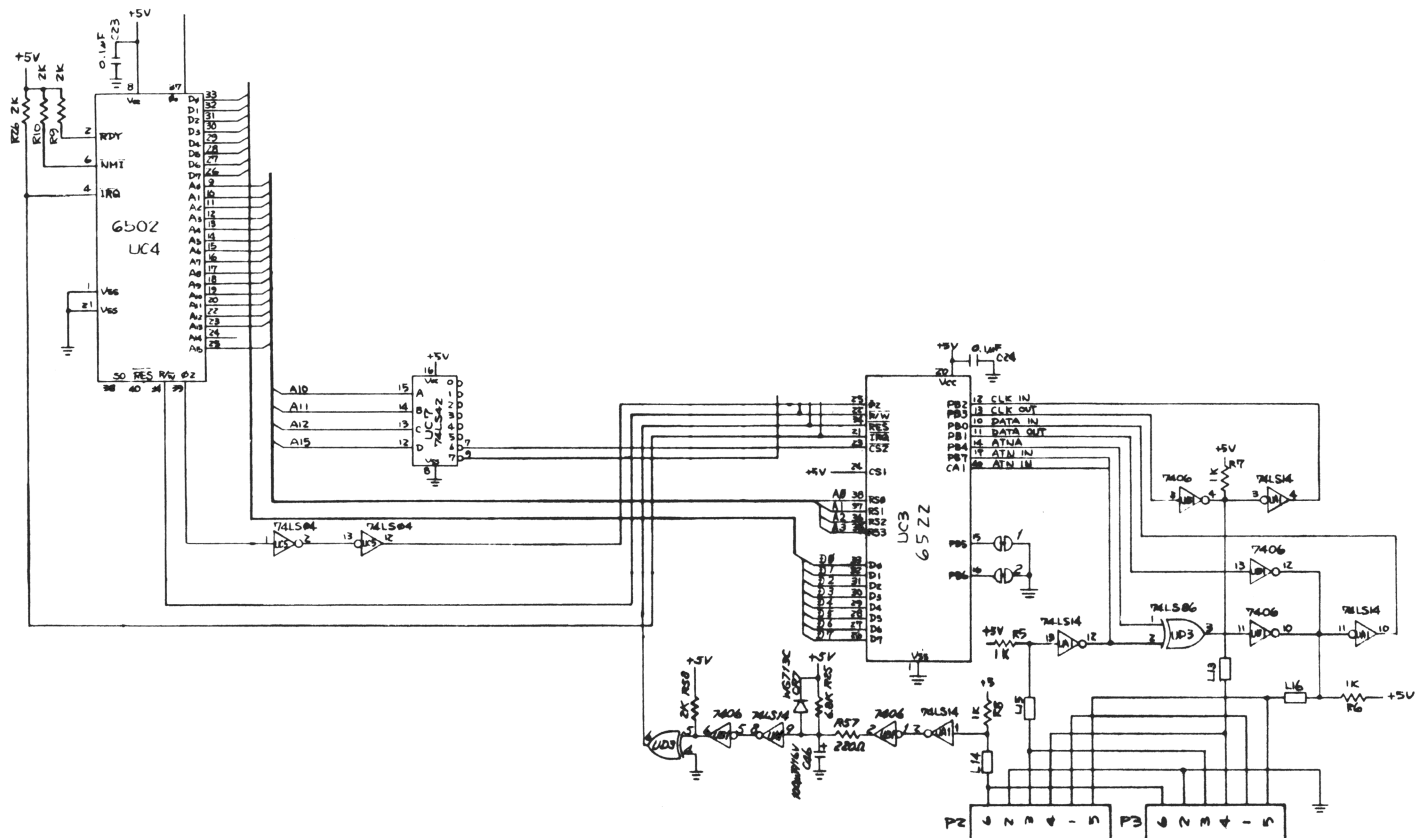


Microprocessor Control of RAM and ROM

UB3 and UB4 are 8192 x 8 bit ROMs that store the Disk Operating System (DOS). UB3 resides at memory locations \$C000-\$DFFF. UB4 resides at memory locations \$E000-\$FFFF. UC5 and UC6 decode the addresses output from the microprocessor when selecting these ROMs.

UB2 is a 2048 x 8 bit RAM. UB2 resides at memory locations \$0000-\$07FF. This memory is used for processor stack operations, general processor housekeeping, user program storage, and 4 temporary buffer areas. UC5, UC6, and UC7 decode the addresses output from the processor when selecting RAM.

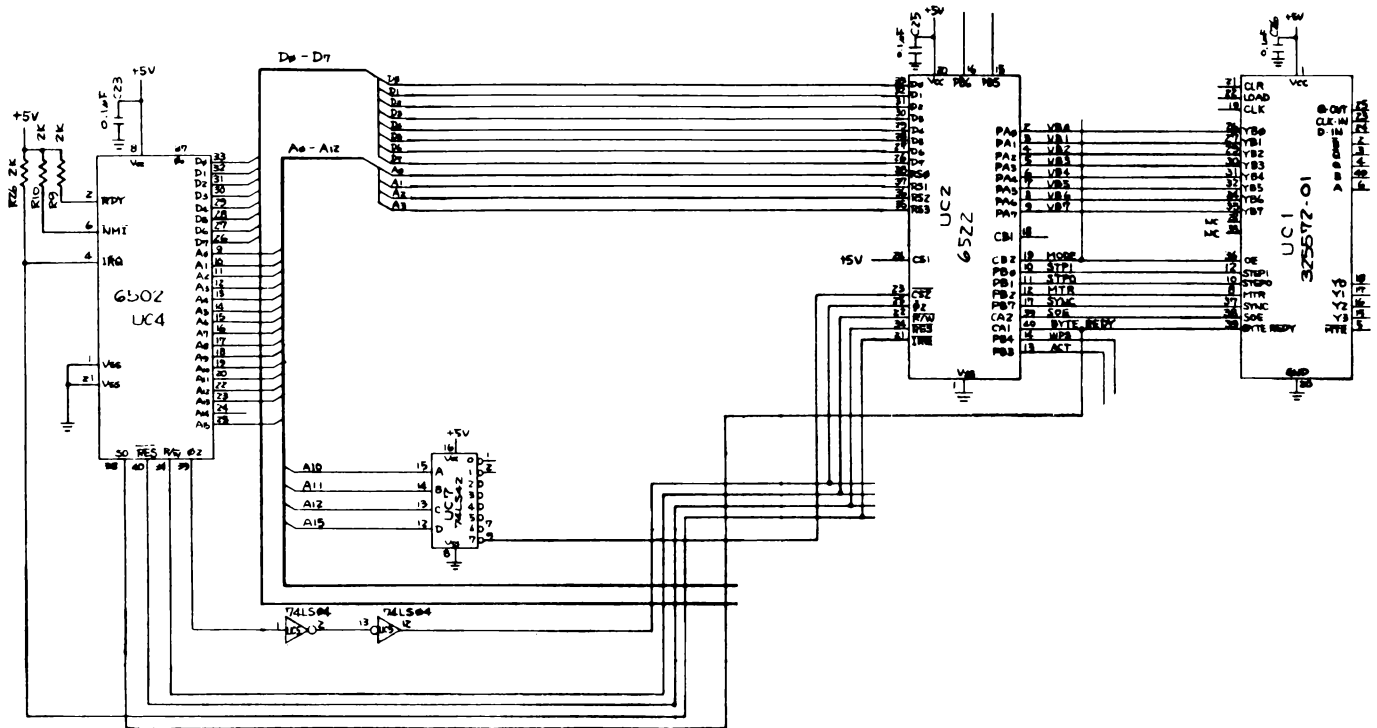
1541 CIRCUIT THEORY



The Serial Interface

UC3 is a 6522 Versatile Interface Adapter (VIA). Two parallel ports, handshake control, programmable timers, and interrupt control are standard features of the VIA. Port B signals (PB0-PB7) control the serial interface driver IC's (UB1 and UA1). CLK and DATA signals are bidirectional signals connected to pins 4 and 5 of P2 and P3. ATN (Attention) is an input on pin 3 of P2 and P3 that is sensed at PB7 and CA1 of UC3 after being inverted by UA1. ATNA (Attention Acknowledge) is an output from PB4 of UC3 which is sensed on the data line pin 5 of P2 and P3 after being exclusively "ored" by UD3 and inverted by UB1. UC3 is selected by UC7 pin 7 going "low" when the proper address is output from the processor. UC3 resides at memory locations \$1C00-\$1C0F.

1541 CIRCUIT THEORY



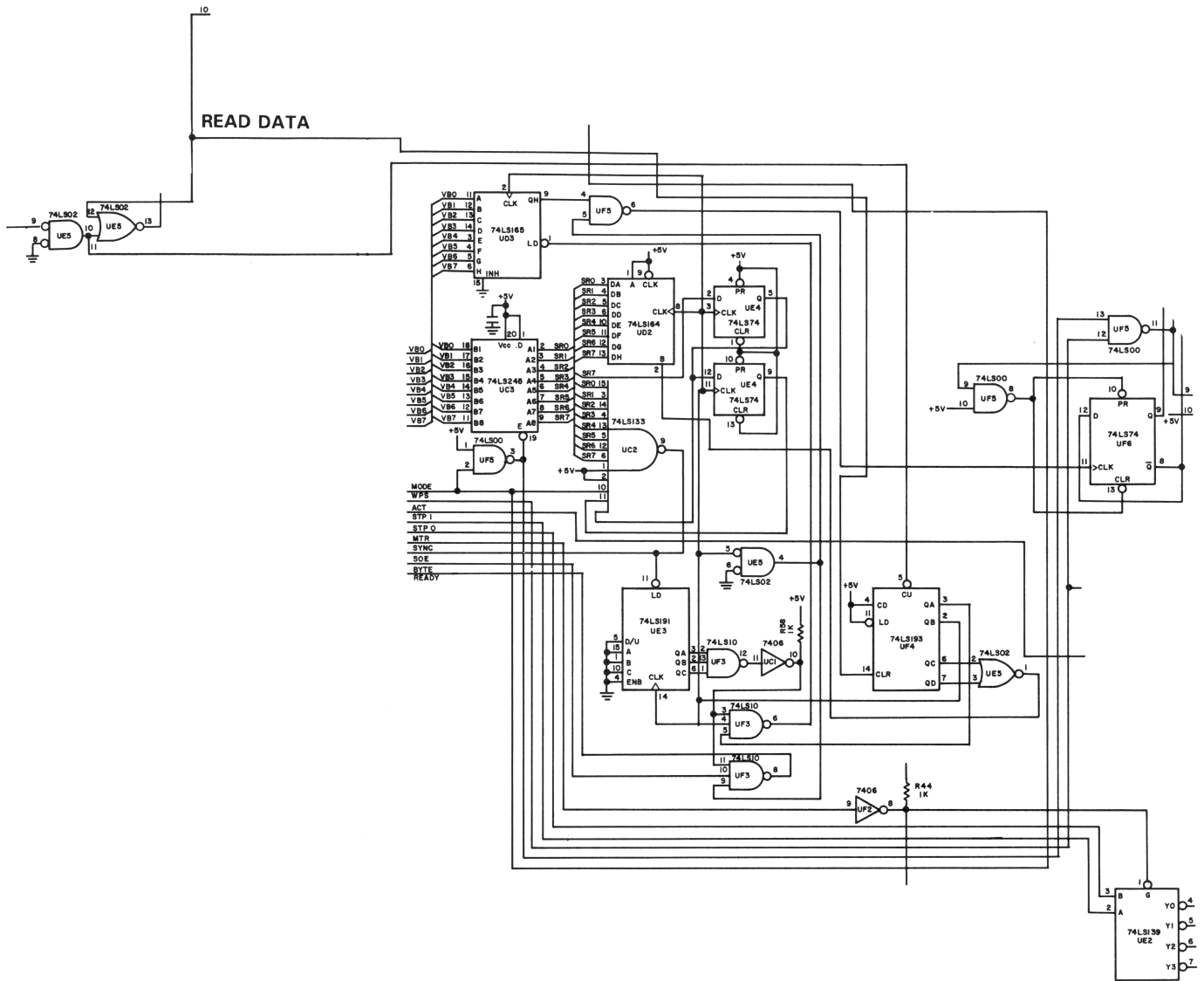
Microprocessor R/W and Motor Control Logic

UC2 is a VIA also. During a write operation the microprocessor passes the data to be recorded to Port A of UC2. The data is then loaded into the PLA parallel port (YB0-YB7). The PLA contains a shift register which converts the parallel data into serial data. The PLA generates signals on pins 2, 3, 4, and 40 which control the write amplifiers during the write operation. During a read operation serial data is received from the read amplifier circuits on D-IN input on pin 24 of the PLA. The PLA shift register converts serial data into parallel data that is latched at the parallel port (YB0-YB7). The microprocessor reads the parallel PLA output by reading Port A of UC2 when BYTE READY on pin 39 goes "low".

The stepper motor is controlled by two outputs on port B of UC2 (STP0, and STP1). A binary four count is developed from these two lines, driving the four phases of the stepper motor. The PLA converts STP0 and STP1 into four outputs that represent one of the four states in the count (Y0,Y1,Y2,Y3). The Spindle motor is controlled by the output MTR of UC2. The PLA inverts this signal. It is then passed to the motor speed control pcb.

UC2 pin 14 is an input that monitors the state of the write protect sensor, and pin 13 is an output that controls the activity light (RED LED). UC7 decodes the addresses output from the processor when selecting UC2. UC2 resides at memory locations \$1800-\$180F.

1541 CIRCUIT THEORY

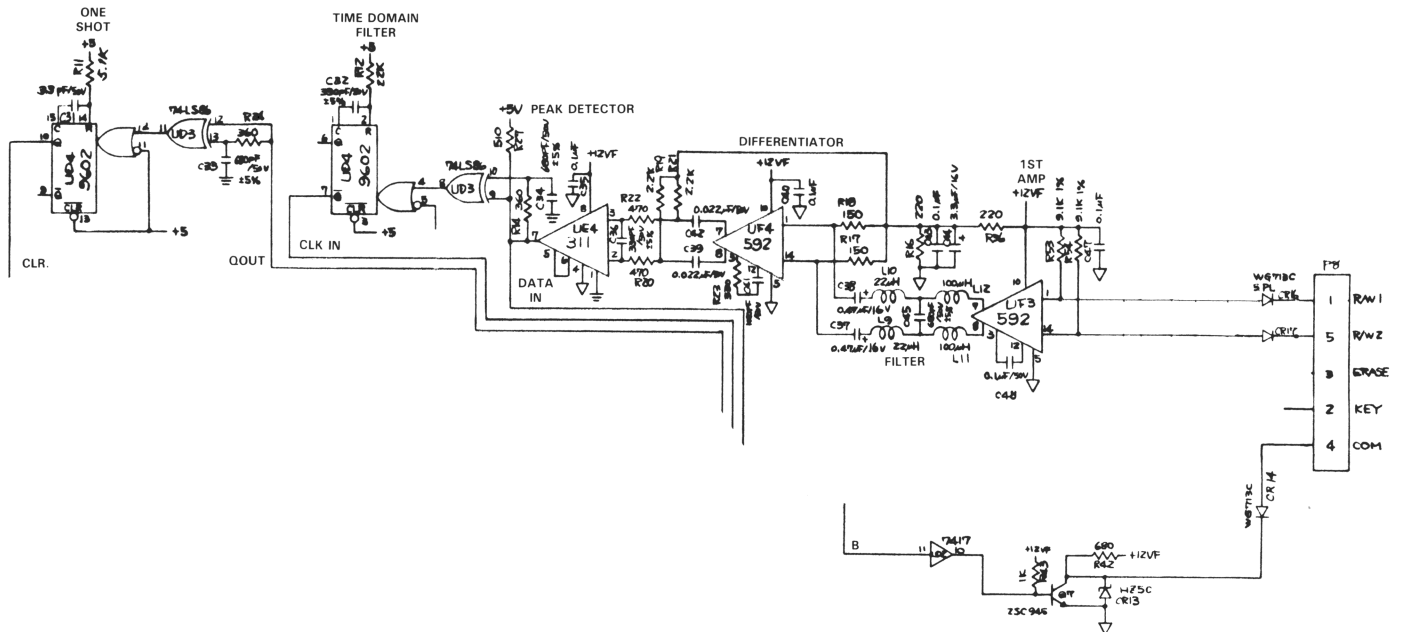


The circuit shown here is from the long board schematic 1540008. This logic was integrated in the PLA (Programmable Logic Array) on the short boards.

Read/Write Control Logic

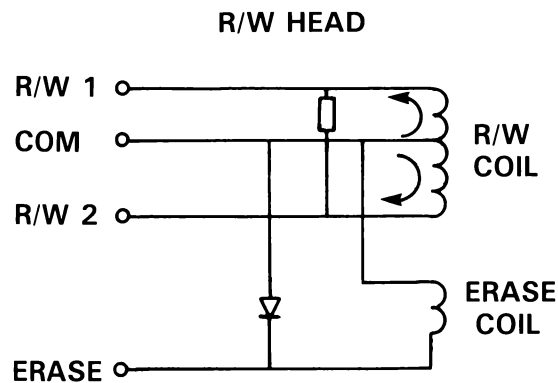
During a write operation, UD3 converts parallel data into serial data. The output on pin 9 is input to 'NAND' gate UF5 pin 4. UF5 outputs the serial data on pin 6 at the clock rate determined by input signal on pin 5. The output clocks the D flip flop UF6. The outputs of UF6, Q and $\overline{Q}$, drive the write amplifiers.

During a read operation, data from the read amplifiers is applied to the CLR input of counter UF4. The outputs, C and D, are shaped by the 'NOR' gate UE5. UE5 outputs the serial data on pin 1, then it is converted to parallel data by UD2. The output of UD2 is latched by UC3. The serial bits are counted by UE3, when 8 bits have been counted, UF3 pin 12 goes "low", UC1 pin 10 goes "high", and UF3 pin 8 goes "low" indicating a byte is ready to be read by the processor. UC2 monitors the parallel output of UD2, when all 8 bits are "1", the output pin 9 goes "low" indicating a sync bit has been read.

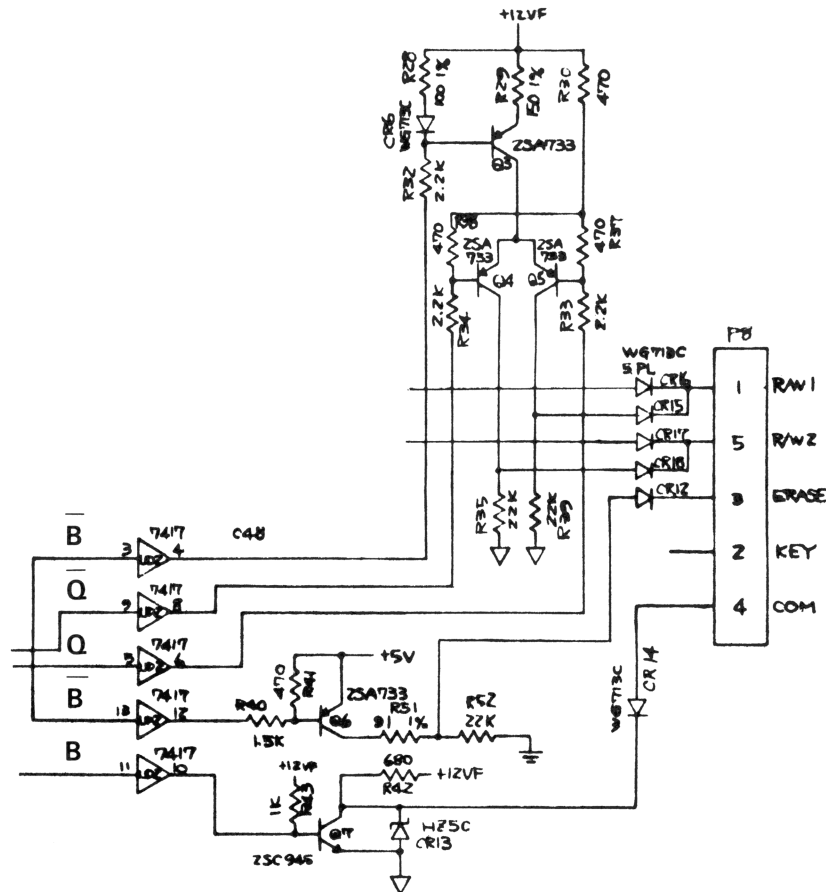


Read Amplifier Circuits

When data is recorded on the disk, a "1" bit is represented on the disk by a change in direction of magnetic flux, caused by a change in direction of current passed through the R/W coil in the R/W head. When a "0" bit is to be recorded, no change in current flow direction occurs, causing the direction of the magnetic flux to remain the same on the disk.



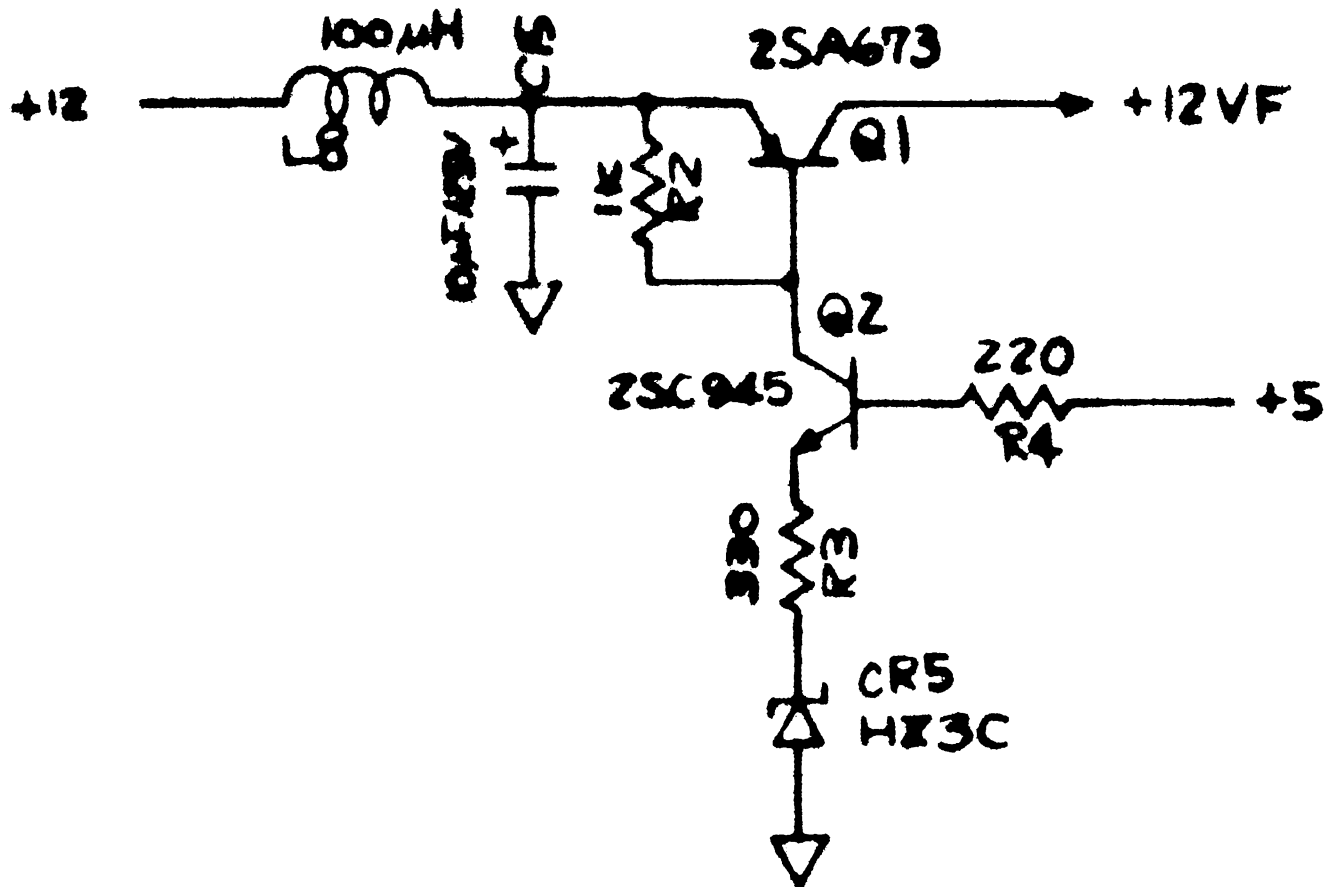
When data is being read from the disk, CEMF is induced into the R/W coil by the magnetic fields on the disk, causing current flow which is detected by the read amplifiers. Current flow through the R/W coil will forward bias either CR16 or CR17, depending on the direction. Q7 and CR14 must be forward biased. The first amplifier UF3, senses this current flow from the R/W coil on one of the inputs and amplifies it. L9, L10, L11, L12 and C45 act as a low pass filter, suppressing noise on the amplified output. UF4 is a differential amplifier which amplifies the difference of the two input signals from the filter section. UE4 is a peak detector. The output of UE4 will pulse "high" when a "1" is read. This signal is the reconstruction of data recorded. The Time Domain Filter, UD4, times out when a "1" bit has been read, so unwanted "1" bits are not added to the actual data. The One Shot UD4 generates the correct data pulse width so the PLA can convert the data to parallel for processor control.



Write Amplifier Circuits

During a write operation, B must be "high". This forward biases Q7 and CR14. If $\overline{B}$ goes "low", Q3 and Q6 become forward biased. If Q goes "low", Q5 and CR15 become forward biased, passing current flow through R/W 1. If $\overline{Q}$ goes "low", Q4 and CR18 become forward biased, passing current flow through R/W 2.

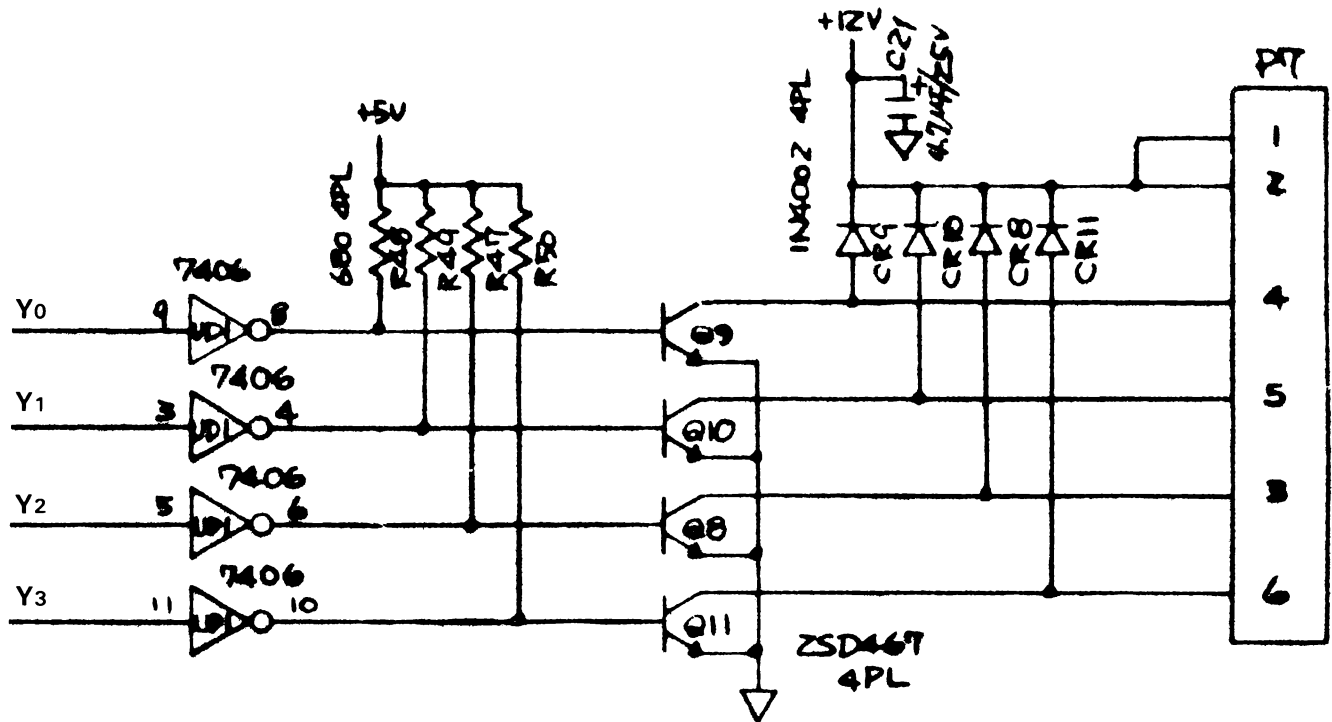
When a write operation occurs, the ERASE coil is energized by forward biasing Q6. This demagnetizes the outer edges of the track, preventing data on one track from bleeding into the next track.



Power Up/Down Write Protection

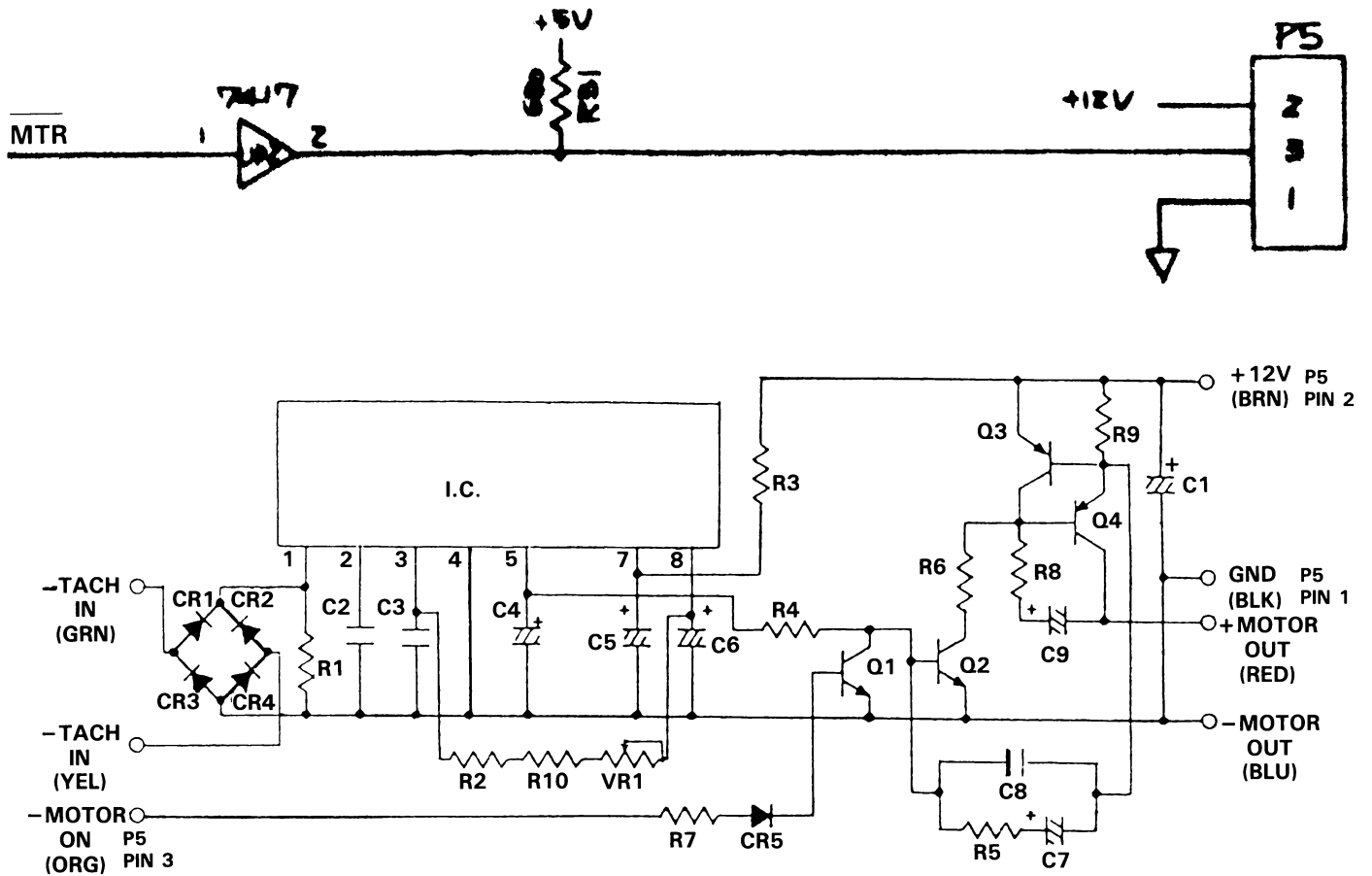
This circuit prevents erroneous data from being written on the disk during power up/down sequences. During a power up, the 12VDC supply is not applied to the R/W coils and amplifier circuits before the processor has control of the logic. During a power down the 12VDC supply is removed from the R/W coils and amplifier circuits before the processor loses control of the logic.

Q1 acts as a series pass transistor, biased to regulate the 12VF output to the R/W coils and Amplifier circuits. Q2 is a feedback amplifier monitoring the 5VDC supply. CR5 develops a precise reference voltage for Q2. L8 and C15 delay the 12VDC supply.



Stepper Motor Control Circuits

Outputs Y0, Y1, Y2, and Y3 from the PLA are inverted by UD1. The outputs of the inverters drive Q8-Q11. The current output from these transistors drive the individual phase coils in the stepper motor and return to the 12VDC supply. CR8-CR11 suppress the CEMF developed by the motor coils.



Spindle Motor Control Circuits

$\overline{\text{MTR}}$ output from the PLA is active "low". This signal is passed, through the current driver UD2, to the motor control PCB. When $\overline{\text{MTR}}$ is "low," Q1 is biased off, and Q2, Q3, and Q4 are biased on, allowing current flow through the spindle motor coil. Attached to the shaft of the spindle motor is an inductive tachometer that generates low level AC voltages, as the motor spins. The output of the tachometer is rectified by CR1-CR4. IC 1 monitors the output of the rectifier and adjusts the bias to Q2, which changes the bias on Q3 and Q4 to regulate motor current for a constant velocity. VR1 is a manual speed adjustment. The speed can be adjusted by watching the 60Hz strobe as the adjustment is made or loading the system test from the diagnostic disc.

The Newtronics Motor Speed PCB is electronically the same as the ALPS Motor Speed PCB, but some of the discrete components have been integrated.

TROUBLESHOOTING GUIDE

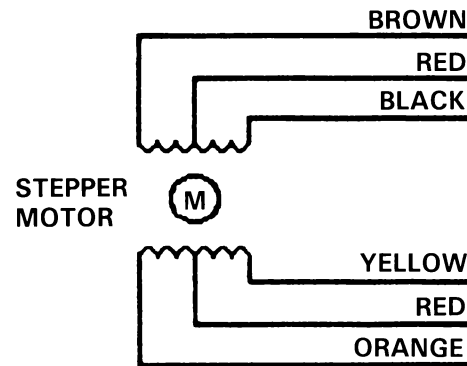
NOTE: Always check for latest ROM/ECO upgrade. If socketed IC is suspected bad, be sure to check socket with ohmmeter.

SYMPTOM:	POSSIBLE SOLUTION:
No LED's on power up.	Is Power cord plugged into wall outlet correctly? Is Power cord plugged into the disk drive correctly? Check line fuse. Check power switch. Check clock on 6502 pin 37. Check + 5 and + 12 volt lines.
Error LED flashes on power up.	Check all RAM and ROM locations.
Error LED stays on all the time.	Check 6502 microprocessor. Check ROMs.
Drive motor runs continuously and red LED stays on.	Check + 12V. Check 6502, logic gates, logic array.
Drive motor runs continuously and red LED stays off.	Check Rom Check drive motor PCB.
Drive motor runs continuously with no red or green LED's.	Check VR2 (5V Regulator). Check Power Transformer.
After the drive warms up the motor runs continuously.	Check 6522s. Check motor control PCB.
Loads programs with red LED flashing.	Check drive speed. Check stepper motor.
Loading is intermittent.	Check ROMs. Check drive alignment.
Does not load when hot or LED flashed 3 times.	Check ROMs.
Searches with LED flashing continuously.	Check ROMs.
Searching with no red LED.	Check 6522s, logic gates, logic array.
Message of 'FILE NOT FOUND' is displayed.	Clean drive head w/alcohol. Check Ø stop adjustment. Check alignment.
Drive fails to read.	Check the 311, 9602, and 592s. There are two + 12 volt sources for stepper output and read circuit, make sure both are good.

TROUBLESHOOTING GUIDE (Continued)

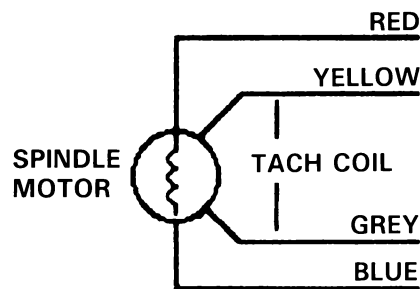
SYMPTOM:	POSSIBLE SOLUTION:
Fails to format disk.	Check components related to connector P7. Check 6522s. Check logic array.
Stepper Motor does not step forward.	Check 6502, 6522s, logic array.
Drive speed will not stabilize.	Check DC motor.
Will not save when the drive heats up.	Check 6502 microprocessor.
Locks-up when loading.	Check serial port components. Check ROM.
Fails the performance test and displays a 21 read error.	Check test diskette. Check Drive Motor.
Fails the performance test and displays a 27 read error.	Check stop adjust.
Passes performance test to track 18 then displays 21 read errors.	Check read/write head.
Passes the performance test but will not load certain programs.	Check stepper motor.

RESISTANCE CHECKS



32 OHMS END TO CENTERTAP

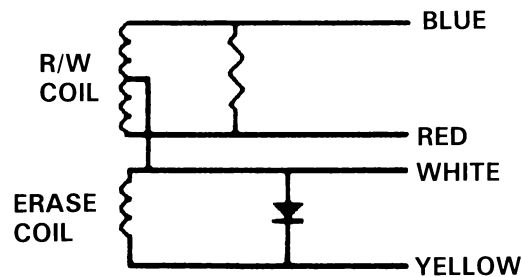
64 OHMS END TO END



MOTOR COIL = 17 OHMS

TACH COIL = 175 OHMS AT REST

TACH COIL = 135 – 190 OHMS IN MOTION



R/W END TO END = 32.4 OHMS

R/W END TO CENTERTAP = 16.3 OHMS

ERASE COIL END TO END = 10.5 OHMS

CASEWORK/ACCESSORY PARTS LIST

1540/1541	TOP CASE (IVORY)	C 1540014-01
1540/41	BOTTOM CASE (IVORY)	C 1540015-00
1541	TOP CASE (BROWN)	C 1540014-03
1541	BOTTOM CASE (BROWN)	C 1540015-01
	SHIELD COVER	1540013-01
	LED ASSEMBLY	1540003
	SELF ADHESIVE FOOT	C 950150-02
	1540 NAMEPLATE	C 1540016-01
	1541 NAMEPLATE	C 1540052-01
	POWER CORD ASSEMBLY	C 903508-04
	6 PIN DIN CABLE	C 1515001-01
	USER'S MANUAL	C 1540031-03
	DEMO DISKETTE	C 1540024-01

NOTE: All 1540 Disk Drives were manufactured prior to any changes to the board and used the Alps drive exclusively.

1541 MODEL IDENTIFICATION

- | | |
|-----------------------|---|
| PCB Assy # 1540008-01 | <ul style="list-style-type: none"> • Schematic # 1540001 • Original "Long" Board • Has 4 discreet 2114 RAMs • ALPS Drive only |
| PCB Assy # 1540048 | <ul style="list-style-type: none"> • Schematic # 1540049 • Referred to as the CR board • Changed to 2048 x 8 bit RAM pkg. • A 40 pin Gate Array is used |
| -01 | <ul style="list-style-type: none"> • Alps Drive |
| -03 | <ul style="list-style-type: none"> • Newtronics Drive |
| PCB Assy # 250442-01 | <ul style="list-style-type: none"> • Schematic # 251748 • Termed the 1541 A • Just one jumper change to accommodate both types of drive |
| PCB Assy # 250446-01 | <ul style="list-style-type: none"> • Schematic # 251748 (See Notes) • Termed the 1541 A-2 • Just one jumper change to accommodate both types of drive |

NOTE: The simplest way to identify which drive you have is by the door assembly. Alps is made with a "push down" door and Newtronics has a "flipper" type door.

DEVICE NUMBER CHANGE

The 1540 and 1541 drives are shipped from the factory set for device # 8. The channel can be hardware altered to # 9, 10, or 11 by two methods:

1) Refer to appropriate board layout drawing for the location of the pads provided for this purpose. The device change pads must be CUT to alter the channel number and re-soldered if another change is needed later.

2) The preferred method to alter the device number is to lift certain pins of the 6522 chip. The I.C. should always be socketed, so removal of the chip is simple. Once removed, the proper pin can be carefully bent to eliminate it from the circuit. If another change is needed, simply remove the I.C. and re-install with the pin back in place.

PCB ASSY #	LOCATION	DEVICE #	LIFT PIN
1540008	UAB1	9	15
1540048,251748	UC3	10	16
and 251834	UC3	11	15 and 16

DEVICE NUMBER
PAD LOCATION

PARTS LIST

PCB ASSEMBLY #1540001

PLEASE NOTE:

Commodore part numbers are provided for reference only and do not indicate the availability of parts from Commodore. Industry standard parts (Resistors, Capacitors, Connectors) should be secured locally. Approved cross-references for TTL chips, Transistors, etc. will be available in manual form through the Service Department in November of 1984. Unique or non-standard parts will be stocked by Commodore and are indicated on the parts list by a "C".

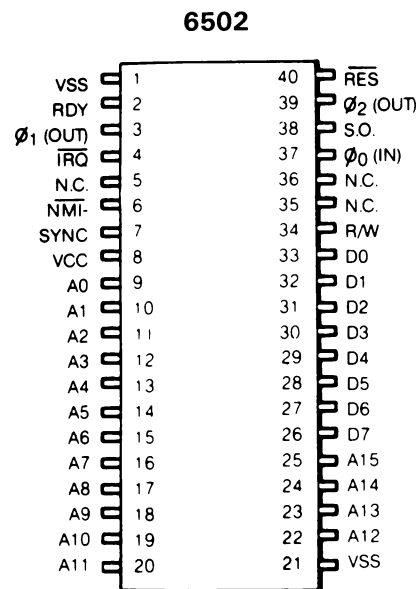
INTEGRATED CIRCUITS			DIODES		
UAB1	6522 VIA	C 901437-01	CR1	Bridge, 1.5A, 50V	900756-01
UAB2	2114 RAM	901471-01	CR2	Rectifier, IN4002	900750-02
UAB3	2114 RAM	901471-01	CR3	Bridge, 4A, 50V	900755-02
UAB4	ROM \$C000-\$DFFF	C 325302-01	CR4	Rectifier, IN4002	900750-02
UAB5	ROM \$E000-\$FFFF (1540)	C 325303-01	CR5	Zener, 3.3V, 500mW, +/- 5%	
UAB5	ROM \$E000-\$FFFF (1541)	C 901229-07		HZ3C-2	325505-01 sub:
UB2	2114 RAM	901471-01		HZ4A-1	325505-02 sub:
UB3	2114 RAM	901471-01		IN5226B	900948-06
UCD4	6522 VIA	C 901437-01	CR6-11	Signal, WG713C	900850-05 sub:
UCD5	6502 CPU	C 901435-01		IN4148	900850-01
UG3	9602	901510-01	CR12	Zener, 5.1V, 500mW, +/- 5%	
UH4	LM311	901523-04		HZ5C-2	325506-01 sub:
UH5	NE 592	901523-08		IN5231	900948-11
UH7	NE 592	901523-08	CR13-16	Rectifier, IN4002	900750-02
TTL			CR17, 18	Signal, WG713C	900850-05 Sub:
				IN4148	900850-01
UB6	74LS04	901521-02	RESISTORS — All Values are in ohms- 1/4 W		
UB7	74LS00	901521-01	5% unless noted otherwise.		
UB8	74LS42	901521-17	R1, 2	330	
UC1	74LS14	901521-30	R3	47	
UC2	74LS133	901521-15	R4	220	
UC3	74LS245	901521-45	R5	330	
UC6	74177	901522-03 sub:	R6	1K	
	74LS197	901521-54	R7	22K	
UC7	7402	901522-32	R8	91, Metal Oxide 1/4W, 1%	
UD1	7406	901522-06	R9	680	
UD2	74LS164	901521-28	R10	22K	
UD3	74LS165	901521-12	R11	1K	
UE2	74LS139	901521-18	R12, 13	9.1K, Metal Oxide 1/4W, 1%	
UE3	74LS191	901521-40	R14, 15	2.2K	
UE4	74LS74	901521-06	R16, 17	220	
UE5	74LS02	901521-21	R18, 19	150	
UE7	74LS193	901521-26	R20	330	
UF2	7406	901522-06	R21-23	2K	
UF3	74LS10	901521-24	R24	510	
UF4	74LS193	901521-26	R25	360	
UF5	74LS00	901521-01	R26	5.1K	
UF6	74LS74	901521-06	R27, 28	470	
UG2	74LS86	901521-32	R29	22K	
UG4	7417	901522-01 sub:	R30	360	
	7407	901522-30	R31-34	1K	
TRANSISTORS			R35, 36	100	
Q1	2SA673 PNP	902720-01	R37	330	
Q2, 3	2SC945 NPN	902671-01 sub:	R38	2K	
	2SC1815 NPN	902693-01	R39-42	680	
Q4, 7	2SD467 NPN	902679-01 sub:	R43	6.8K	
	2SC2120 NPN	902682-01	R44	1K	
Q8-11	2SA733 PNP	902717-01 sub:	R45	220	
	2SA1015 PNP	902744-01	R46	100K	

PARTS LIST PCB ASSEMBLY #1540001 (Continued)

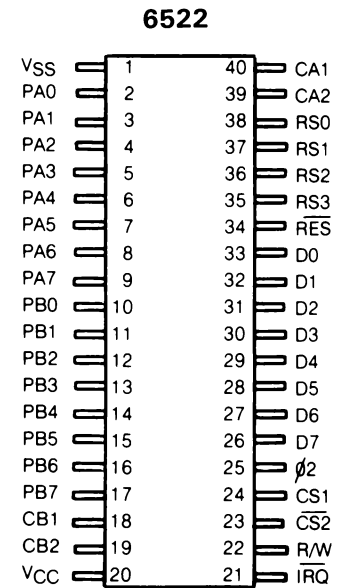
C - Indicates Commodore Stocked Part Number

RESISTORS (Continued)				CAPACITORS (Continued)			
— All values are in ohms- 1/4 W 5% unless noted otherwise.							
R47	470			C51	Electrolytic	6800 μ F	25V
R48	1.5K			C52	Electrolytic	10000 μ F	16V
R49	100, Metal Oxide 1/4W, 1%			C53-55	Ceramic	.1 μ F	50V
R50	470			C56	Electrolytic	100 μ F	16V
R51, 52	2.2K			C57	Ceramic	.1 μ F	50V
R53	22K			C58, 59	Ceramic	.022 μ F	16V
R54	150, Metal Oxide 1/4W, 1%			C60, 61	Ceramic	.1 μ F	50V
R55	470			C62	Tantalum	4.7 μ F	25V
R56	2.2K			C63	Tantalum	1 μ F	35V
R57	470			C64	Ceramic	.033 μ F	25V
R58	1K			C65	Electrolytic	220 μ F	25V
R59	220			MISCELLANEOUS			
CAPACITORS				L1	Coil Inductor 2.2 μ H		
C1	Electrolytic	1 μ F	25V	L2-6	Ferrite Bead (2743005112)		
C2	Electrolytic	47 μ F	16V	L7	Coil Inductor 100 μ H		
C3	Ceramic	.1 μ F	50V	L8	Coil Inductor 22 μ H		
C4	Electrolytic	1 μ F	25V	L9, 10	Coil Inductor 100 μ H		
C5	Electrolytic	47 μ F	16V	L11	Coil Inductor 22 μ H		
C6-9	Ceramic	.1 μ F	50V	L12-16	Ferrite Bead (2743005112)		
C10	Ceramic	68pF	50V	VR1	Voltage Reg 12V, 1.5A (LM340K-12 TO-3)		
C11	Ceramic	.1 μ F	50V	VR2	Voltage Reg 5V, 1.2A (LM340K-5 TO-3) Sub:		
C12	Tantalum	10 μ F	25V		5V, 3A (LM323)		
C13, 14	Ceramic	.1 μ F	50V	Y1	Crystal 16MHz	900556-02	
C15	Tantalum	.47 μ F	16V + / - 20%		Shield Box	4022048-01	
C16	Ceramic	680 pF	50V + / - 5%		Shield Cap	4022047-01	
C17-22	Ceramic	.1 μ F	50V	CONNECTORS			
C23	Tantalum	3.3 μ F	25V	P1	Header Assy, 3.96 Pitch, 4Pin	(Mole 5271-04A)	
C24	Tantalum	.47 μ F	16V + / - 20%	P2, 3	6 Pin Din,		
C25	Ceramic	.1 μ F	50V		(Hoshidenki TSC4460-01-101	C 903361-01	
C26	Ceramic	1000pF	50V	P4, 5	Header Assy, 2.5 Pitch, 3Pin	(Molex 3022-03A)	
C27	Ceramic	680pF	50V + / - 5%	P6	Header Assy, 2.5 Pitch, 15Pin	(Molex 3022-15A)	
C28	Ceramic	330pF	50V + / - 5%	P7	Header Assy, 2.5 Pitch, 6Pin	(Molex 3022-06A)	
C29-32	Ceramic	.1 μ F	50V	P8	Header Assy, 2.5 Pitch, 4Pin	(Molex 5048-04AG)	
C33	Ceramic	33pF	50V				
C34-48	Ceramic	.1 μ F	50V				
C49	Ceramic	330pF	50V + / - 5%				
C50	Ceramic	680pF	50V + / - 5%				

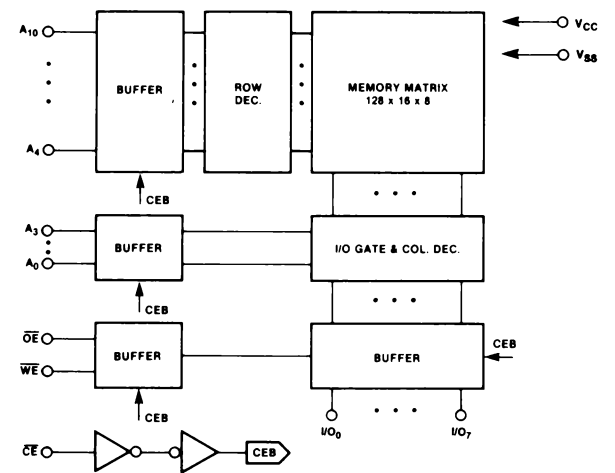
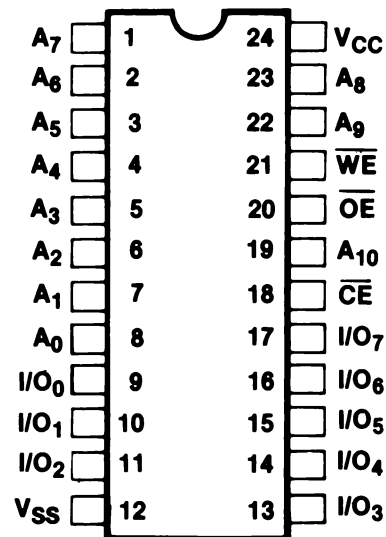
PIN CONFIGURATIONS:



901435-01
Microprocessor



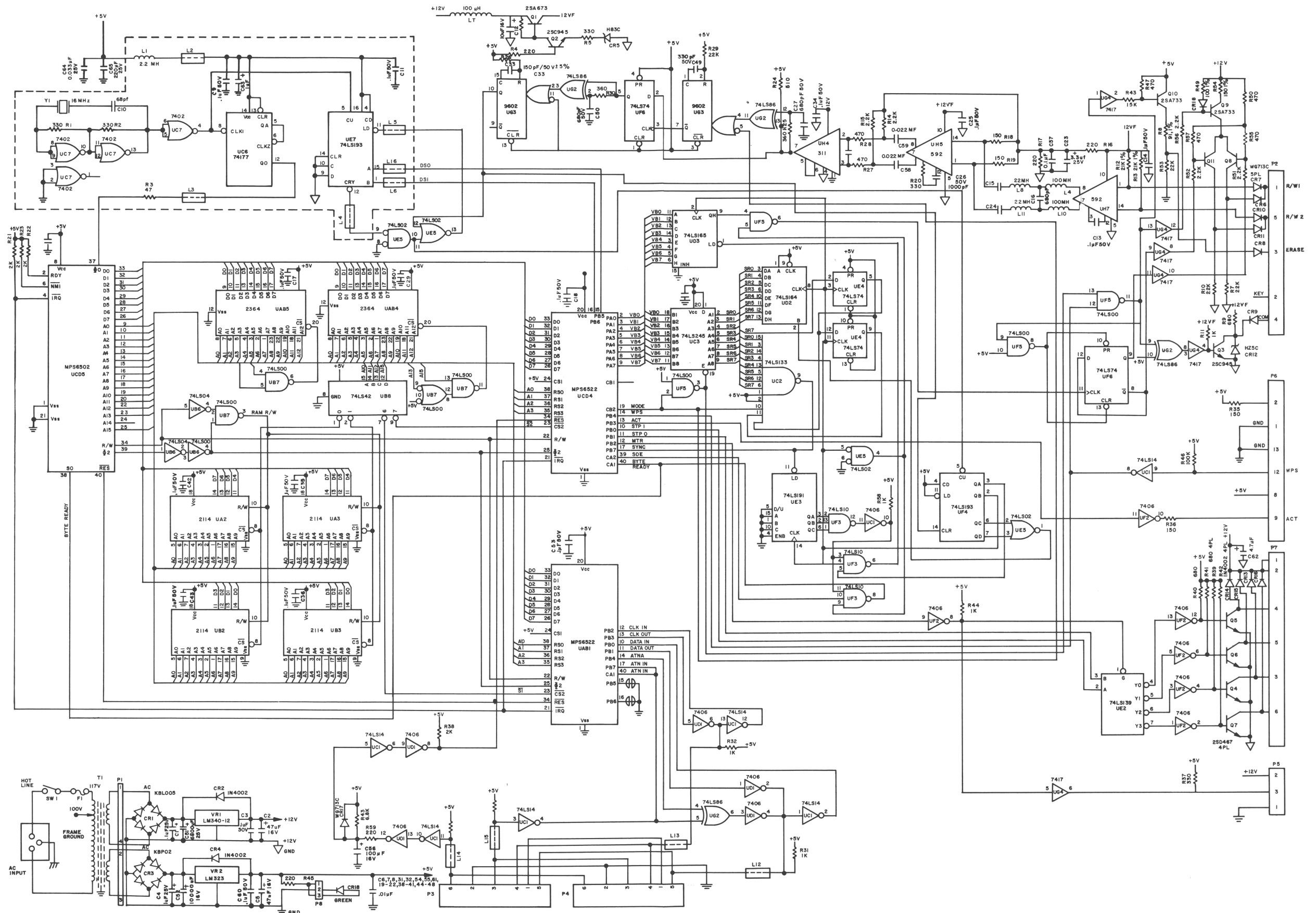
901437-01
Versatile Interface
Adapter (VIA)



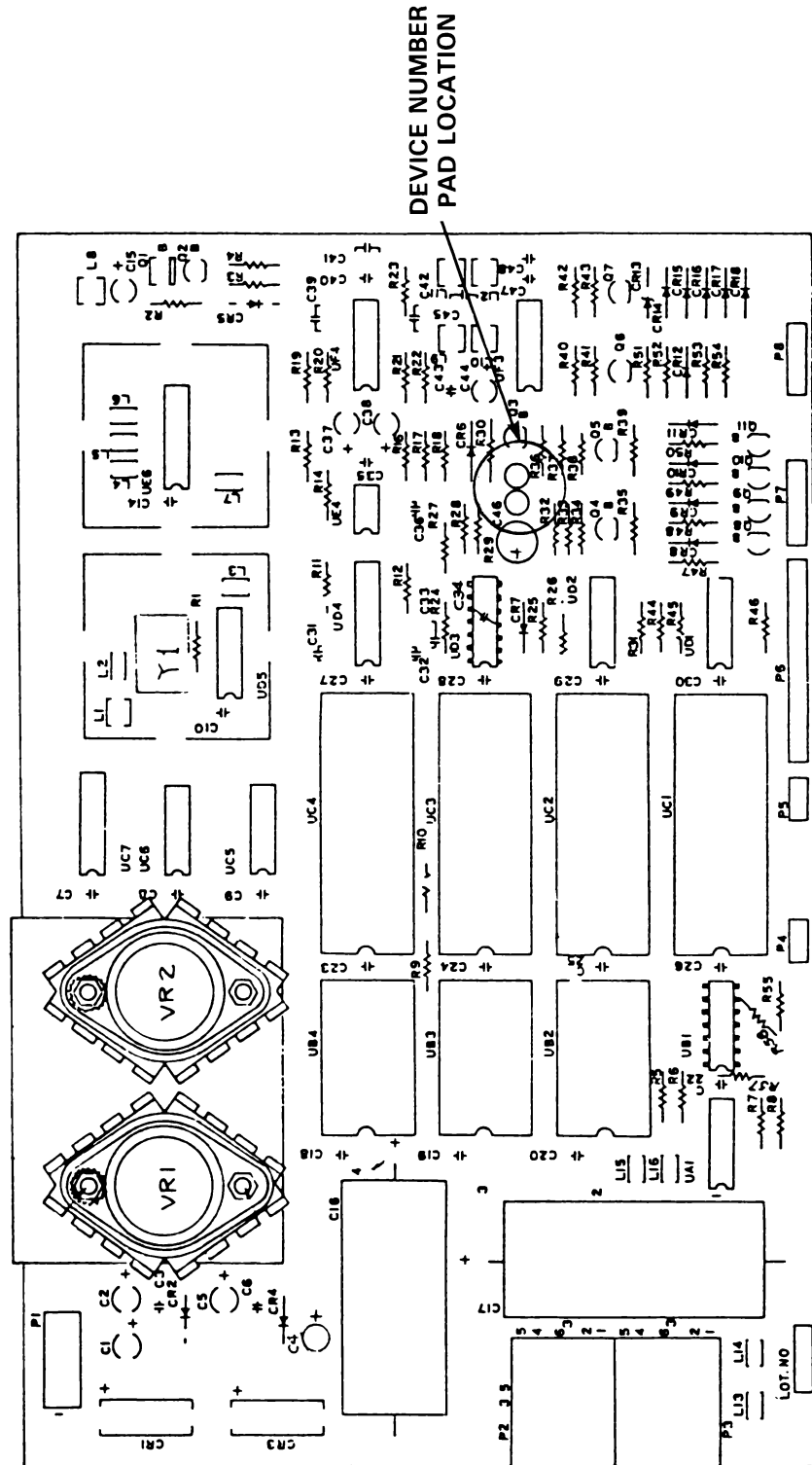
2048 X 8 STATIC RAM

Approved Replacements	
Mitsubishi	M58725P
Toshiba	TMM2016P
Hitachi	HM6116LP-4/AP-20

SCHEMATIC #1540008



PCB ASSY #1540048 BOARD LAYOUT



PARTS LIST

PCB ASSEMBLY #1540048

PLEASE NOTE:

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INTEGRATED CIRCUITS				DIODES		
UB2	TMM2016P RAM	325502-03	sub:	CR1	Bridge, 1.5A, 50V	900756-01
	M58725P RAM	325502-01	sub:	CR2	Rectifier, IN4002	900750-02
	HM6116P4	325502-08		CR3	Bridge, 1.5A, 50V	900756-01
UB3	ROM \$C000-\$DFFF	C 325302-01		CR4	Rectifier, IN4002	900750-02
UB4	ROM \$E000-\$FFFF	C 901229-05		CR5	Zener, 3.3V, 500mW, +/- 5%	
UC1	GATE ARRAY	C 325572-01			HZ3C-2	325505-01 sub:
UC2	6522 VIA	C 901437-01			HZ4A-1	325505-02 sub:
UC3	6522 VIA	C 901437-01			IN5226B	900948-06
UC4	6502 CPU	C 901435-01		CR6, 7	Signal, WG713C	900850-05 sub:
UD4	9602	901510-01			IN4148	900850-01
UE4	LM311	901523-04		CR8-11	Rectifier, IN4002	900750-02
UF3	NE 592	901523-08		CR12	Signal, WG713C	900850-05 sub:
UF4	NE 592	901523-08			IN4148	900850-01
TTL				CR13	Zener, 5.1V, 500mW, +/- 5%	
UA1	74LS14	901521-30			HZ5C-2	325506-01 sub:
UB1	7406	901522-06	sub:		IN5231	900948-11
	7707	251749-01		CR14-18	Signal, WG713C	900850-05 Sub:
UC5	74LS04	901521-02	sub:		IN4148	900850-01
	7713	251749-02	sub:	RESISTORS — All Values are in ohms- 1/4 W		
	74LS14	901521-30	sub:	5% unless noted otherwise.		
	7414	901522-19		R1	47	
UC6	74LS00	901521-01		R2	1K	
UC7	74LS42	901521-17		R3	330	
UD1	7406	901522-06		R4	220	
UD2	7417	901522-01	sub:	R5-8	1K	
	7407	901522-30		R9, 10	2K	
UD3	74LS86	901521-32		R11	5.1K	
UD5	74LS197	901521-54	sub:	R12	22K	
	74177	901522-03		R14	360	
UE6	74LS193	901521-26		R16	220	
TRANSISTORS				R17, 18	150	
Q1	2SA673	PNP	902720-01	R19	2.2K	
Q2	2SC945	NPN	902671-01	R20	470	
	2SC1815	NPN	902693-01	R21	2.2K	
Q3-6	2SA733	PNP	902717-01	R22	470	
	2SA1015	PNP	902744-01	R23	330	
Q7	2SC945	NPN	902671-01	R24	360	
	2SC1815	NPN	902693-01	R25	6.8K	
Q8-11	2SD467	NPN	902679-01	R26	2K	
	2SC2120	NPN	902682-01	R27	510	
				R28	100 @ 1%	
				R29	150 @ 1% (ALPS)	
					137 @ 1% (NEWTRONICS)	
				R30	470	
				R31	680	
				R32-34	2.2K	
				R35	22K	
				R36	220	
				R37, 38	470	
				R39	22K	

PARTS LIST PCB ASSEMBLY #1540048 (Continued)

C - Indicates Commodore Stocked Part Number

RESISTORS (Continued)		CAPACITORS (Continued)	
— All Values are in ohms- 1/4 W 5% unless noted otherwise.			
R40	1.5K	C39	Ceramic .022μF, 16V
R41	470	C40	Ceramic .1μF, 50V
R42	680	C41	Ceramic 1000pF, 50V
R43	1K	C42	Ceramic .022μF, 16V
R44	100K	C43	Ceramic .1μF, 50V
R45, 46	150	C44	Tantalum 3.3μF, 25V
R47-50	680	C45	Ceramic 680pF, 50V +/- 5%
R51	91 @ 1% (ALPS) 56 @ 1% (NEWTRONICS)	C46	Electrolytic 100μF, 16V
R52	22K	C47,48	Ceramic .1μF, 50V
R53, 54	9.1K @ 1%	MISCELLANEOUS	
R55	220	L1	Coil Inductor 2.2μH
R57	220	L2-7	Ferrite Bead (2743005112)
R58	2K	L8	Coil Inductor 100μH
CAPACITORS		L9,10	Coil Inductor 22μH
C1	Electrolytic 1μF, 25V	L11,12	Coil Inductor 100μH
C2	Electrolytic 47 μF, 16V	L13-16	Ferrite Bead
C3	Ceramic .1μF, 50V	VR1	Voltage Reg 12V, 1.5A (LM340K-12 TO-3)
C4	Electrolytic 1μF, 25V	VR2	Voltage Reg 5V, 1.2A (LM340K-5 TO-3) Sub: 5V, 3A (LM323)
C5	Electrolytic 47μF, 16V	Y1	Crystal Module 16MHz 50ppm (NDK, Tyocom) Sub: 16MHz 100ppm (NDK, Tyocom, Kyocera)
C6-10	Ceramic .1μF, 50V	Shield Box	C 4022048-01
C11	Tantalum 1μF, 35V	Shield Cap	C 4022047-01
C12	Ceramic .033μF, 25V	CONNECTORS	
C13	Electrolytic 220μF, 10V	P1	Header Assy, 3.96 Pitch, 4Pin (Molex 5271-04A)
C14	Ceramic .1μF, 50V	P2, 3	6 Pin Din, C 903361-01
C15	Tantalum 10μF, 25V	P4, 5	Header Assy, 2.5 Pitch, 3Pin (Molex 3022-03A)
C16	Electrolytic 4700μF, 16V	P6	Header Assy, 2.5 Pitch, 15Pin (Molex 3022-15A)
C17	Electrolytic 6800μF, 25V	P7	Header Assy, 2.5 Pitch, 6Pin (Molex 3022-06A)
C18-20	Ceramic .1μF, 50V	P8	Header Assy, 2.5 Pitch, 4Pin (Molex 5048-04AG)
C21	Tantalum 4.7μF, 25V		
C22-30	Ceramic .1μF, 50V		
C31	Ceramic 33pF, 50V +/- 5%		
C32	Ceramic 330pF, 50V +/- 5%		
C33, 34	Ceramic 680pF, 50V +/- 5%		
C35	Ceramic .1μF, 50V		
C36	Ceramic 330pF, 50V +/- 5%		
C37-38	Tantalum .47μF, 16V		

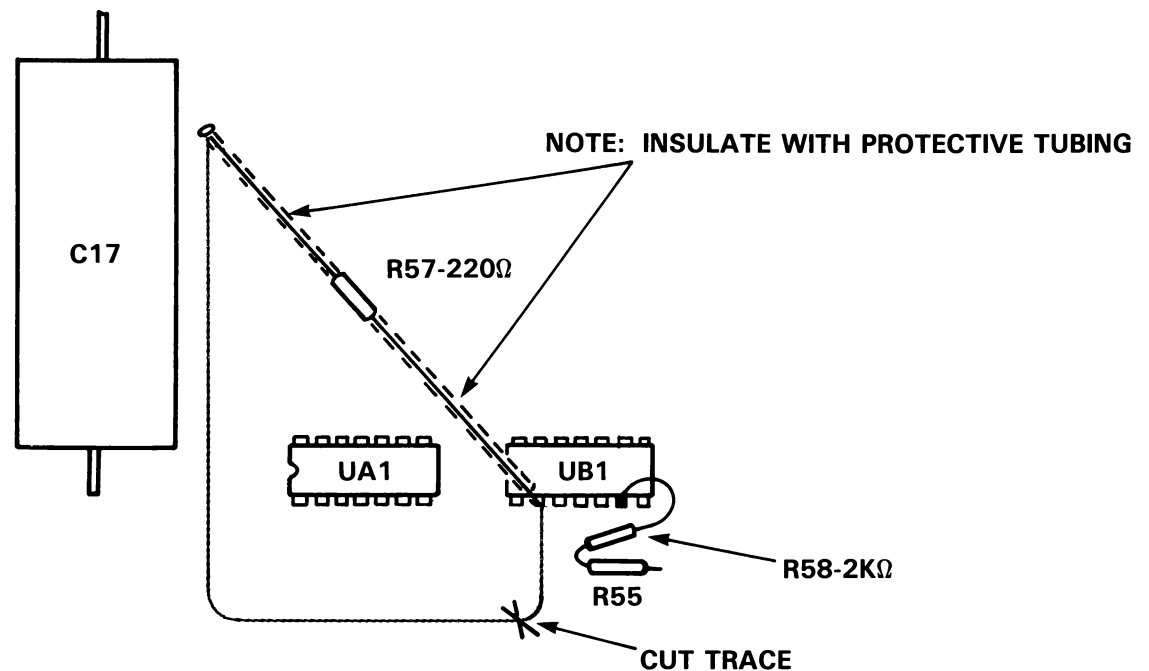
UPGRADE NOTES

PCB Assembly # 1540048 (Schematic 1540049) requires an upgrade to the reset circuit. The upgrade consists of:

1) COMPONENT CHANGES:

R25	Change To	6.8K ohm,	1/4 W, +/ - 5%
R57	Add	220 ohm,	1/4 W, +/ - 5%
R58	Add	2K ohm,	1/4 W, +/ - 5%

2) CUT circuit trace near UA1 (See diagram)



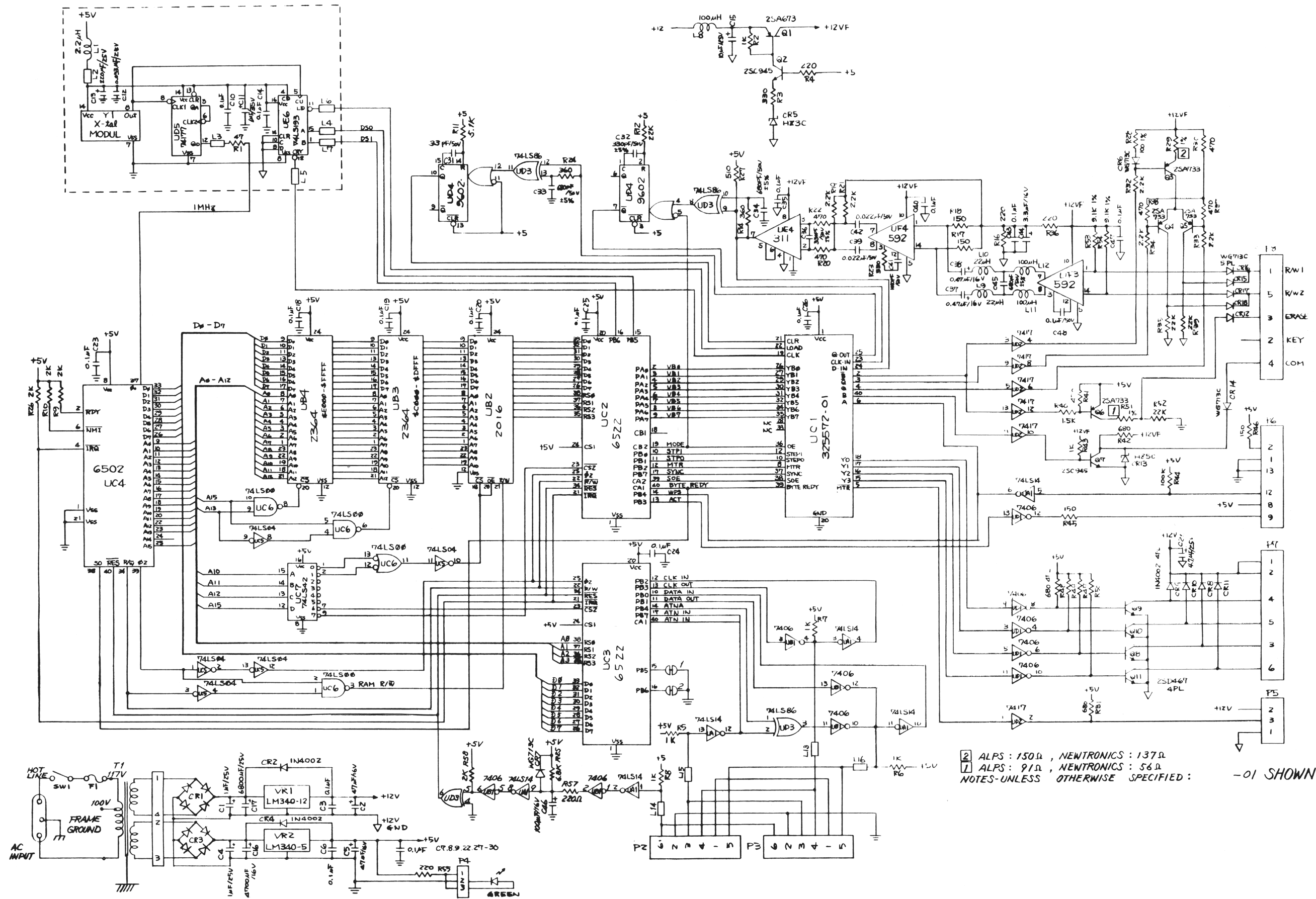
3) INSTALL JUMPER WIRES:

BETWEEN:

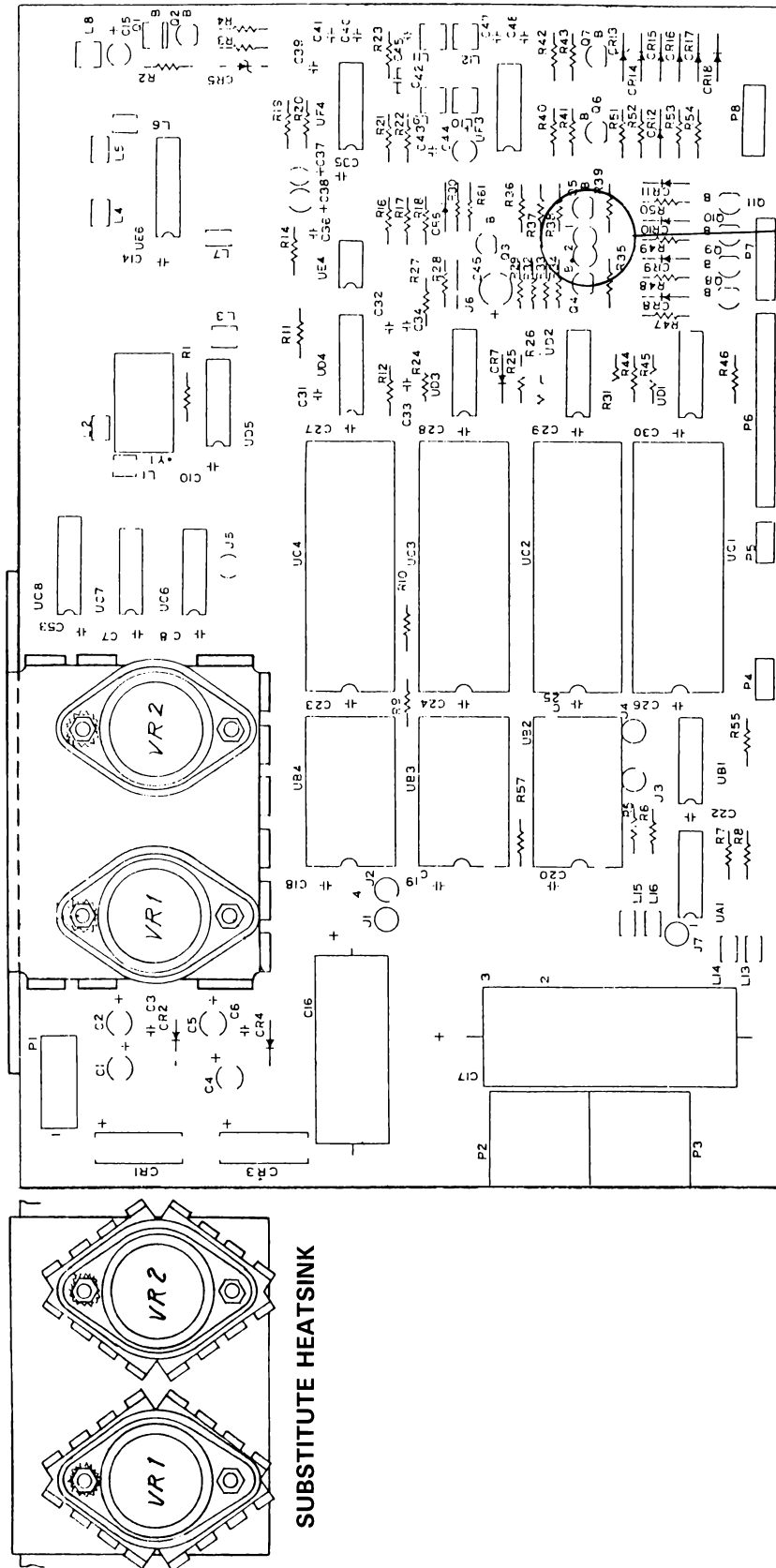
- UA1 pin 8 and UB1 pin 5
- UA1 pin 9 and CR7 Anode
- UB1 pin 6 and UD3 pin 5 on BACK of board

4) CUT circuit trace between CR7 Anode and UD3 pin 5 on BACK of board.

SCHEMATIC #1540049



PCB ASSEMBLY #250442/46 BOARD LAYOUT



DEVICE
NUMBER
PAD
LOCATION

PARTS LIST

PCB ASSEMBLY #250442/46

PLEASE NOTE:

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INTEGRATED CIRCUITS				DIODES (Continued)		
UB2	TMM2016P RAM	325502-03	sub:	CR6, 7	Signal, WG713C	900850-05 sub:
	M58725P RAM	325502-01	sub:		IN4148	900850-01
	HM6116P4	325502-08		CR8-11	Rectifier, IN4002	900750-02
UB3	ROM \$C000-\$DFFF	C 325302-01		CR12	Signal, WG713C	900850-05 sub:
UB4	ROM \$E000-\$FFFF	C 901229-05			IN4148	900850-01
UC1	GATE ARRAY	C 325572-01		CR13	Zener, 5.1V, 500mW, +/- 5%	
UC2	6522 VIA	C 901437-01			HZ5C-2	325506-01 sub:
UC3	6522 VIA	C 901437-01			IN5231	900948-11
UC4	6502 CPU	C 901435-01		CR14-18	Signal, WG713C	900850-05 Sub:
UD4	9602 (250442)	901510-01	sub:		IN4148	900850-01
	74LS123 (250446)	901521-49		RESISTORS — All values are in ohms- 1/4 W		
UE4	LM311	901523-04		5% unless noted otherwise.		
UF3	NE 592	901523-08		R1	47	
UF4	NE 592	901523-08		R2	1K	
TTL				R3	330	
UA1	74LS14	901521-30		R4	220	
UB1	7406	901522-06	sub:	R5-8	1K	
	74LS06	901521-73		R9, 10	2K	
UC5	74LS04	901521-02	sub:	R11	5.1K	
	74LS14	901521-30	sub:	R12	22K (250442)	
	7414	901522-19			15K	
UC7	74LS00	901521-01		R14	360 (250446)	
UC8	74LS42	901521-17		R16	220	
UD1	7406	901522-06	sub:	R17, 18	150	
	74LS06	901521-73		R19	2.2K	
UD2	7417	901522-01	sub:	R20	470	
	7407	901522-30		R21	2.2K	
UD3	74LS86	901521-32		R22	470	
UD5	74LS197	901521-54	sub:	R23	330	
	74177	901522-03		R24	360	
UE6	74LS193	901521-26		R25	6.8K	
TRANSISTORS				R26	2K	
Q1	2SA673 PNP	902720-01		R27	510	
Q2	2SC945 NPN	902671-01	sub:	R28	100 @ 1%	
	2SC1815 NPN	902693-01		R29	137 @ 1%	
Q3-6	2SA733 PNP	902717-01	sub:	R30	470	
	2SA1015 PNP	902744-01		R31	680	
Q7	2SC945 NPN	902671-01	sub:	R32-34	2.2K	
	2SC1815 NPN	902693-01		R35	22K	
Q8-11	2SD467 NPN	902679-01	sub:	R36	220	
	2SC2120 NPN	902682-01		R37, 38	470	
DIODES				R39	22K	
CR1	Bridge, 1.5A, 50V	900756-01		R40	1.5K	
CR2	Rectifier, IN4002	900750-02		R41	470	
CR3	Bridge, 1.5A, 50V	900756-01		R42	680	
CR4	Rectifier, IN4002	900750-02		R43	1K	
CR5	Zener, 3.3V, 500mW, +/- 5%			R44	100K	
	HZ3C-2	325505-01	sub:	R45, 46	150	
	HZ4A-1	325505-02	sub:	R47-50	680	
	IN5226B	900948-06		R51	56 @ 1%	
				R52	22K	
				R53, 54	10K	
				R55	220	
				R57	220	
				R61	13 @ 1%	

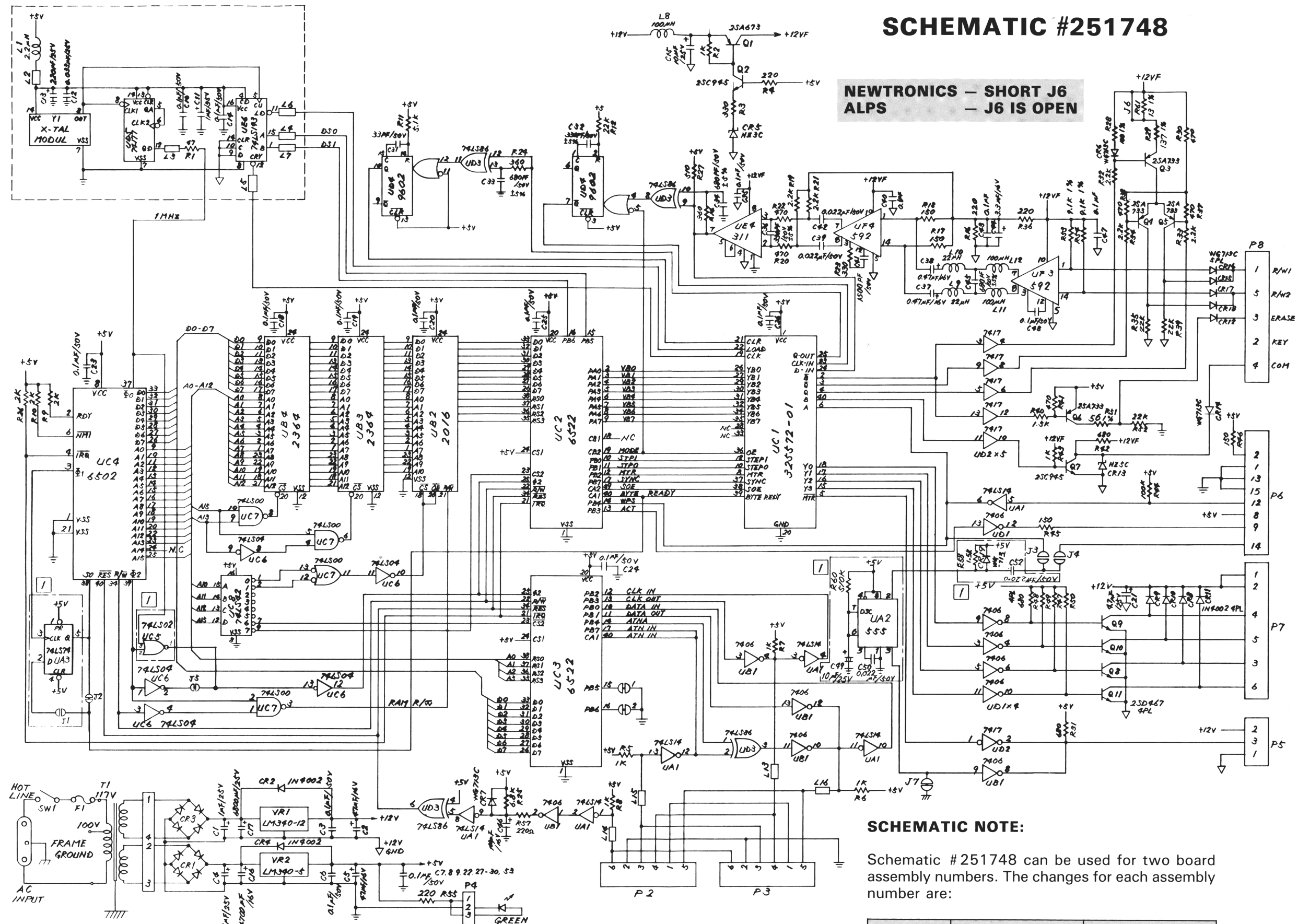
PARTS LIST PCB ASSEMBLY #250442/46 (Continued)

C - Indicates Commodore Stocked Part Number

CAPACITORS			MISCELLANEOUS	
C1	Electrolytic	1 μ F, 25V	L1	Coil Inductor 2.2 μ H
C2	Electrolytic	47 μ F, 16V	L2-7	Ferrite Bead (2743005112)
C3	Ceramic	.1 μ F, 50V	L8	Coil Inductor 100 μ H
C4	Electrolytic	1 μ F, 25V	L9,10	Coil Inductor 22 μ H
C5	Electrolytic	47 μ F, 16V	L11,12	Coil Inductor 100 μ H
C6-10	Ceramic	.1 μ F, 50V		
C11	Tantalium	1 μ F, 35V	VR1	Voltage Reg 12V, 1.5A (LM340K-12 TO-3)
C12	Ceramic	.033 μ F, 25V	VR2	Voltage Reg 5V, 1.2A (LM340K-5 TO-3) Sub: 5V, 3A (LM323)
C13	Electrolytic	220 μ F, 10V		
C14	Ceramic	.1 μ F, 50V	Y1	Crystal Module 16MHz 50ppm (NDK, Tyocom) Sub: 16MHz 100ppm (NDK, Tyocom, Kyocera)
C15	Tantalium	10 μ F, 25V		
C16	Electrolytic	4700 μ F, 16V		
C17	Electrolytic	6800 μ F, 25V		
C18-20	Ceramic	.1 μ F, 50V		
C21	Tantalium	4.7 μ F, 25V		
C22-30	Ceramic	.1 μ F, 50V		
C31	Ceramic	33pF, 50V + / - 5%		
C32	Ceramic	330pF, 50V + / - 5%		
C33,34	Ceramic	680pF, 50V + / - 5%		
C35	Ceramic	.1 μ F, 50V		
C36	Ceramic	330pF, 50V + / - 5%		
C37,38	Tantalium	.47 μ F, 16V		
C39	Ceramic	.022 μ F, 16V		
C40	Ceramic	.1 μ F, 50V		
C41	Ceramic	1000pF, 50V		
C42	Ceramic	.022 μ F, 16V		
C43	Ceramic	.1 μ F, 50V		
C44	Tantalium	3.3 μ F, 25V		
C45	Ceramic	680pF, 50V + / - 5%		
C46	Electrolytic	100 μ F, 16V		
C47,48	Ceramic	.1 μ F, 50V		
			CONNECTORS	
			P1	Header Assy, 3.96 Pitch, 4Pin (Molex 5271-04A)
			P2, 3	6 Pin Din, C 903361-01
			P4, 5	Header Assy, 2.5 Pitch, 3Pin (Molex 3022-03A)
			P6	Header Assy, 2.5 Pitch, 15Pin (Molex 3022-15A)
			P7	Header Assy, 2.5 Pitch, 6Pin (Molex 3022-06A)
			P8	Header Assy, 2.5 Pitch, 4Pin (Molex 5048-04AG)

SCHEMATIC #251748

NEWTRONICS — SHORT J6
ALPS — J6 IS OPEN



SCHEMATIC NOTE:

Schematic # 251748 can be used for two board assembly numbers. The changes for each assembly number are:

Component	PCB Assy #250442	PCB Assy #250446
UD4	9602	74LS123
R12	22K	15K
C31	33pF	Eliminated

PCB Assembly #250442 is shown here.

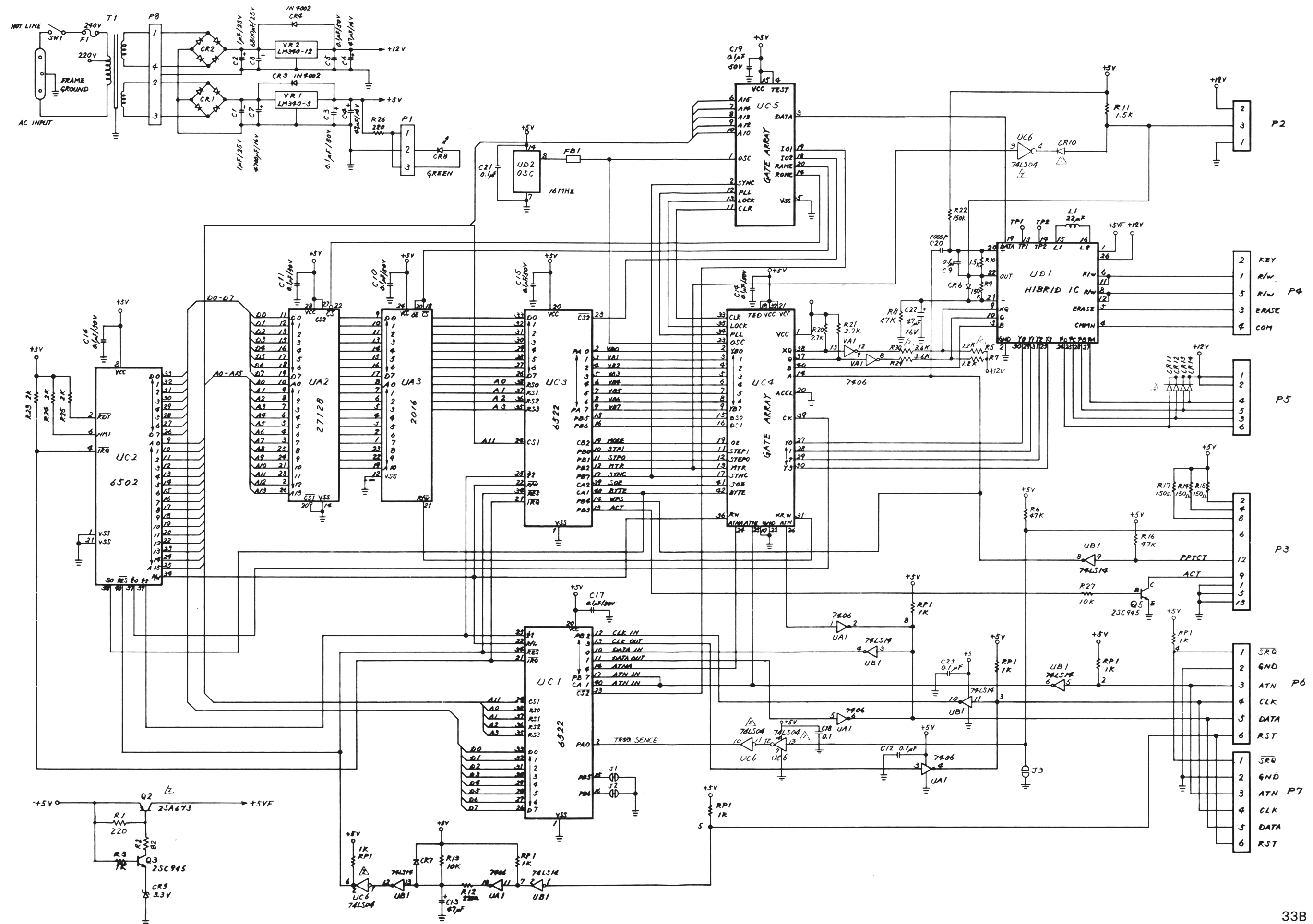
PARTS LIST

PCB ASSEMBLY #250448

PLEASE NOTE:

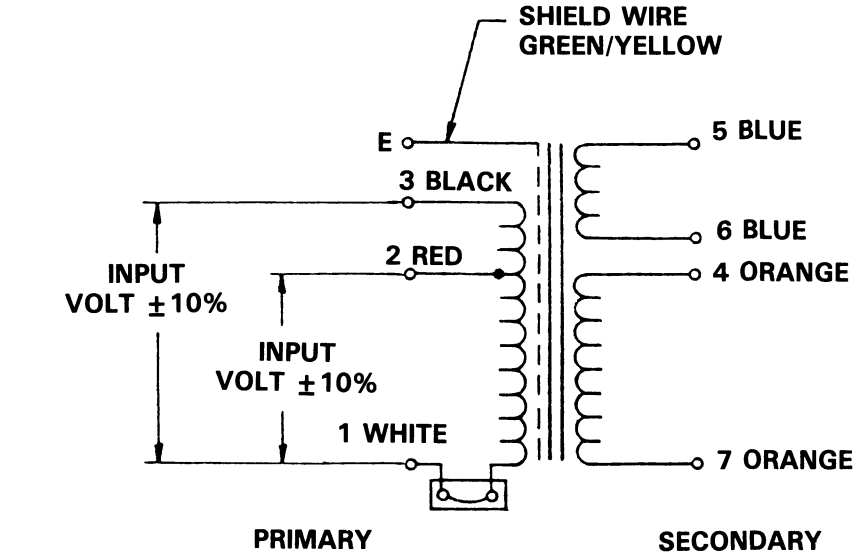
Commodore part numbers are provided for reference only and do not indicate the availability of parts from Commodore. Industry standard parts (Resistors, Capacitors, Connectors) should be secured locally. Approved cross-references for TTL chips, Transistors, etc. are available in manual form through the Service Department, order part #314000-01. Unique or non-standard parts will be stocked by Commodore and are indicated on the parts list by a "C". Vendor Name and part number have been provided for your convenience in ordering custom or unique parts.

INTEGRATED CIRCUITS		RESISTORS (Continued)	
UA1	7406	R20,21	2.7K
UA2	27128 EPROM C 251968-01	R22	150K
UA3	TMM2016P RAM C 325502-03	R23,24,25	2K
UB1	74LS14	R26	220
UC1	6522 VIA C 901437-01	R27	10K
UC2	6502 CPU C 901435-01	R29,30	3.6K†
UC3	6522 VIA C 901437-01		
UC4	Gate Array 40 Pin C 251828-01 sub:		
	Gate Array 42 Pin 251828-02		
UC5	Gate Array 20 Pin C 251829-01		
UC6	74LS04		
UD1	R/W HYBRID C 251853-01* sub:		
	R/W HYBRID C 251853-02*		
*NOTE:	Diode and resistor variations required when using the -01 or -02 R/W HYBRID are noted throughout the parts list.		
TRANSISTORS		CAPACITORS	
Q2	2SA673 PNP	C1,2	Electrolytic 1μF, 25V
Q3,5	2SC945 NPN	C3	Ceramic .1μF, 50V
		C4	Electrolytic 47μF, 16V
		C5	Ceramic .1μF, 50V
		C6	Electrolytic 47μF, 16V
		C7	Electrolytic 4700μF, 16V
		C8	Electrolytic 6800μF, 25V
		C9-12	Ceramic .1μF, 50V
		C13	Electrolytic 47μF, 16V
		C14-19	Ceramic .1μF, 50V
		C20	Ceramic 1000pF, 50V
		C21	Ceramic .1μF, 50V
		C22	Electrolytic 47μF, 16V
		C23	Ceramic .1μF, 50V
DIODES		MISCELLANEOUS	
CR1,2	Bridge, 1.5A, 50V (KBP-005)	L1	Coil Inductor 22μH
CR3,4	Rectifier, IN4002	FB1	Ferrite Bead
CR5	Zener, 3.3V, 500mW, +/-5% HZ4A-1	VR1	Voltage Reg 5V 1.2A (LM340K-5 to -3) Sub:
CR6,7,9	Signal, 1N4148		5V 3A (LM323)
CR11-14*	Rectifier, IN4002	VR2	Voltage Reg 12V 1.5A (LM340K-12 to -3)
		UD2	Crystal Module 16 MHz C 325566-01
		RP1	Resistor Pack 1K, 8 Pin
RESISTORS — All are carbon 1/4 watt, 5% unless noted otherwise.		CONNECTORS	
R1	220	P1,2	Header Assy, 2.5 Pitch, 3 Pin (Molex 3022-03A)
R2	82K	P3	Header Assy, 2.5 Pitch, 15 Pin (Molex 3022-15A)
R3	1K	P4	Header Assy, 2.5 Pitch, 4 Pin (Molex 5048-04AG)
R5	1.2K*	P5	Header Assy, 2.5 Pitch, 6 Pin (Molex 3022-06A)
R6	47K	P6,7	6 Pin Din, C 903361-01
R7	1.2K*	P8	Header Assy, 3.96 Pitch, 4 Pin (Molex 5271-04A)
R8	47K		
R9	150K		
R10,11	1.5K		
R12	220		
R13	10K		
R14,15	150		
R16	47K		
R17	150		

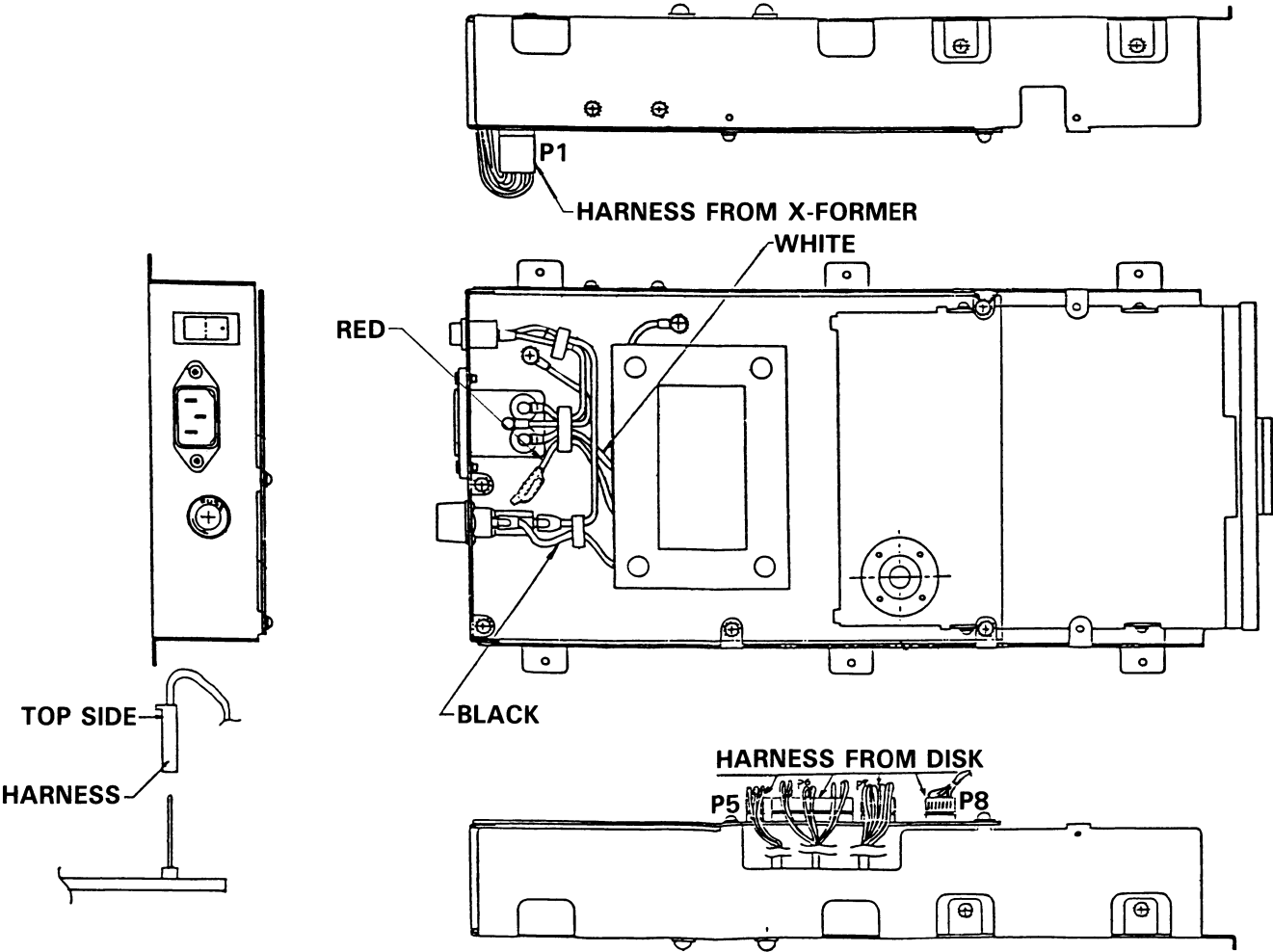


1540/1541 POWER SUPPLY ASSEMBLY
PARTS LIST

FUSE HOLDER	903614-01
ROCKER SWITCH	904509-01
POWER CNNCT FILTER	903467-03 sub:
	325552-01
FUSE, SLOW BLO, 250V, 1.0A	903556-16
POWER TRANSFORMER	1540009-02



TRANSFORMER



ASSEMBLY DRAWING

ALPS DRIVE ASSEMBLIES 1540/1541

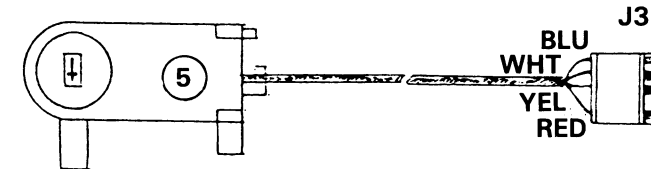
PARTS LIST

32551901 Alps Drive (Black)

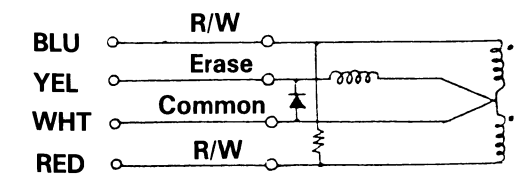
32551902 Alps Drive (Brown)

- | | |
|--|---|
| <p>① 31410001 ALP DOOR/HUB ASSEMBLY
1-Door Assy w/Spring
2-Hub/Collet Assy
3-Arm Support Assy</p> <p>② 31410101 ALP LEFT DISK GUIDE ASSEMBLY
1-Diskette Guide
2-LED Assy w/Harness
3-Write Protect Assy</p> <p>③ 31410201 ALP RIGHT DISK GUIDE</p> <p>④ 31410301 ALP FRONT BEZEL (Black)

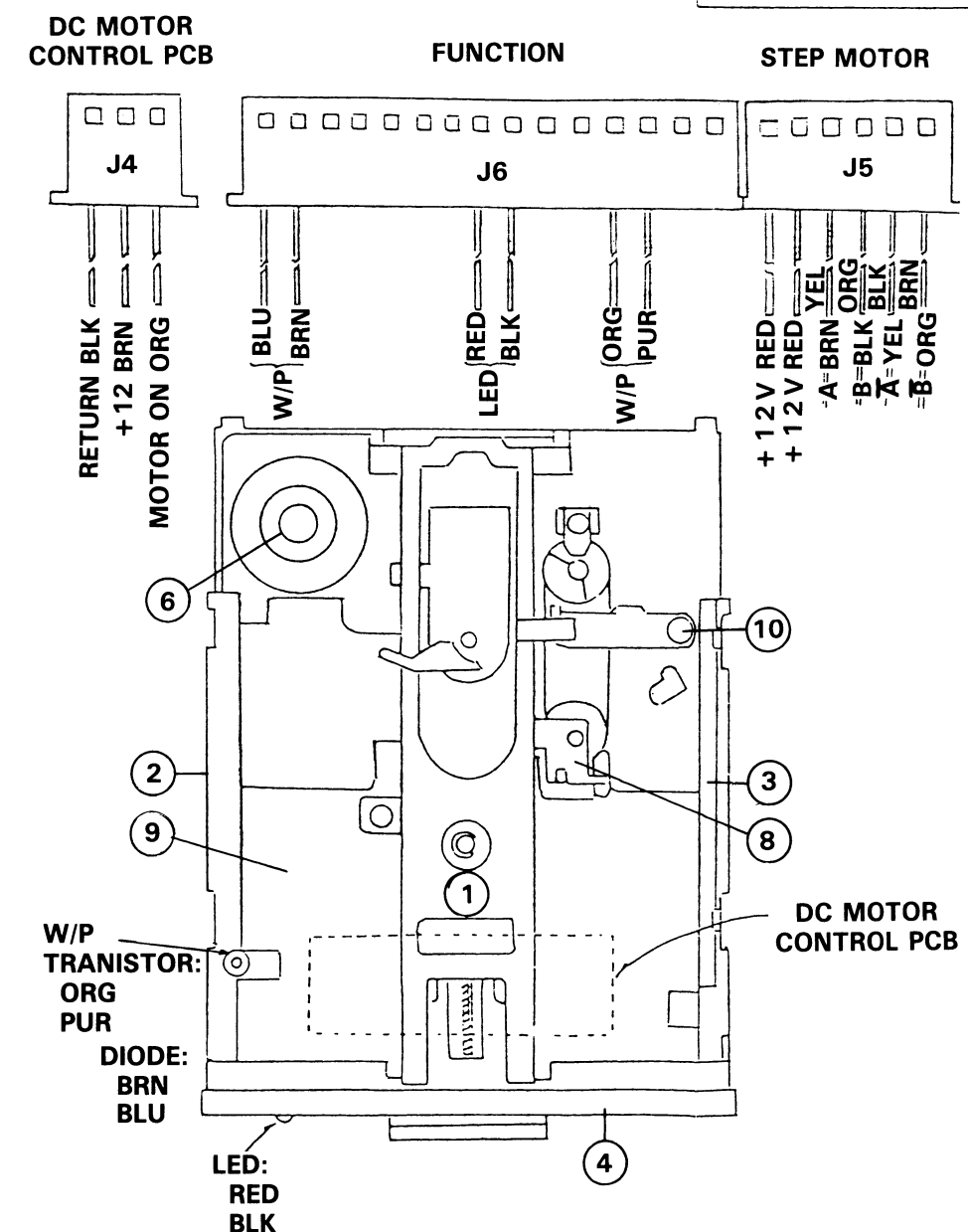
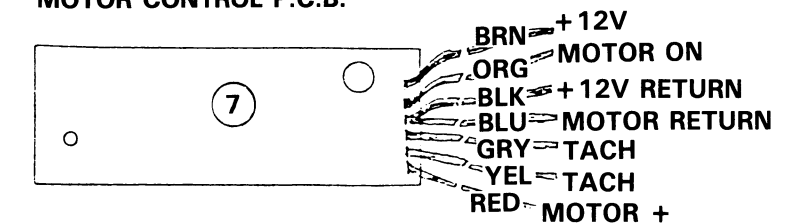
31410302 ALP FRONT BEZEL (Brown)</p> <p>⑤ 31410401 ALP R/W HEAD ASSEMBLY
1-R/W Head
2-Load Arm w/Pad
3-Metal Band</p> <p>31410501 ALP STEPPER MOTOR ASSEMBLY
1-Stepper Motor w/Harness
2-Stepper Pulley</p> <p>⑥ 31410601 ALP D.C. MOTOR</p> | <p>⑦ 31410701 ALP MOTOR CONTROL PCB ASSEMBLY
1-Motor Control PCB
2-Harness Assy</p> <p>⑧ 31410801 ALP TENSION PULLEY ASSEMBLY
1-Pulley Wheel w/Spring
2-Plastic Housing</p> <p>⑨ 31410901 ALP HOUSING/SPINDLE ASSEMBLY
1-Housing Base
2-Spindle Assy
3-L/R Guide Shafts</p> <p>31411001 ALP DRIVE BELT</p> <p>⑩ 31411101 ALP EJECT ASSEMBLY
1-Eject Plate
2-Eject Spring</p> <p>31411201 ALP HARDWARE
1-Assorted Screws
2-Zero Stop Tab</p> <p>31417401 UNIV Replacement Load Pad</p> |
|--|---|



READ/WRITE HEAD ASSEMBLY



MOTOR CONTROL P.C.B.



PARTS LIST

NEWTRONICS DRIVE ASSEMBLY
1541

#251643-01 NEWTRONICS DRIVE (Brown)
#251643-02 NEWTRONICS DRIVE (Dk. Grey)

- ①

314120-01 NWT DOOR/LATCH ASSEMBLY
1-Top Plate w/Latch Spring & Cam
2-Latch Knob w/Allen Screws
- ②

314121-01 NWT HUB FRAME ASSEMBLY
1-Hub/Collet Assy
2-Hub Frame
- ③

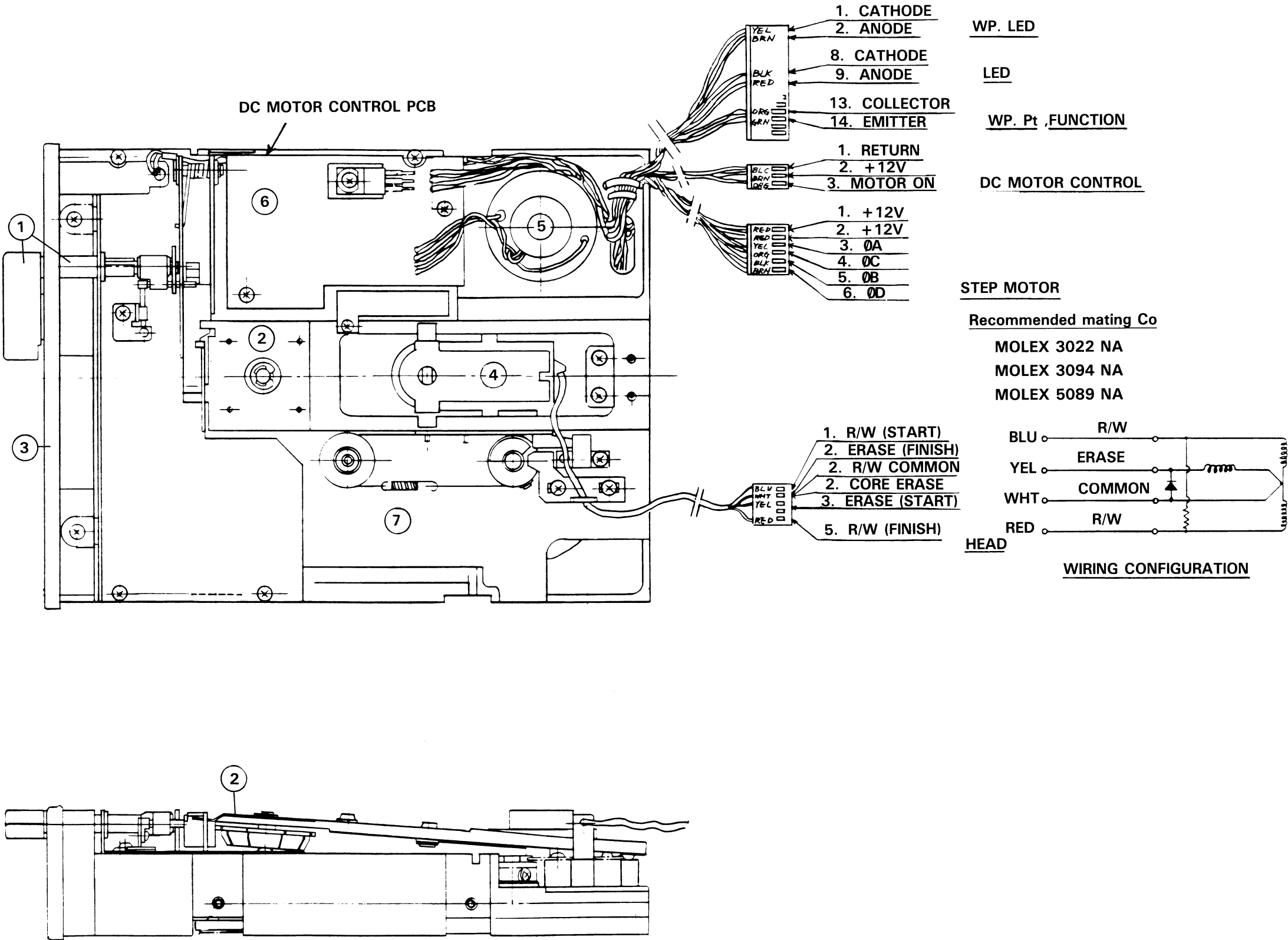
314122-01 NWT FRONT BEZEL ASSEMBLY
1-Front Bezel w/LED & Harness
2-Write Protect Assy
- ④

314123-01 NWT R/W HEAD ASSEMBLY
1-R/W Head
2-Load Arm w/Pad
3-Metal Band
- 314124-01 NWT STEPPER MOTOR ASSEMBLY
1-Stepper Motor w/Harness
2-Stepper Pulley
- ⑤

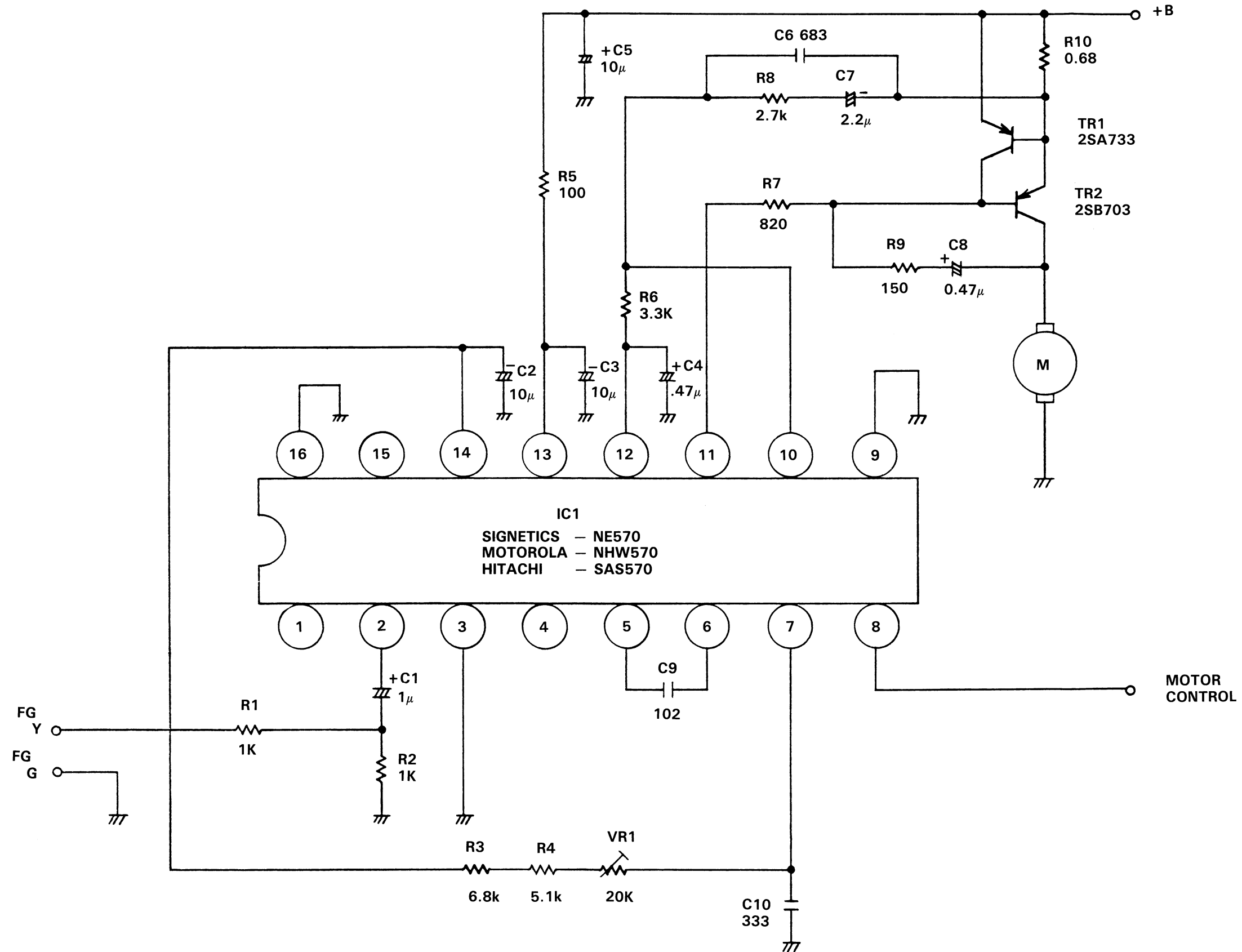
314125-01 NWT D.C. MOTOR
- ⑥

314126-01 NWT MOTOR CONTROL PCB ASSEMBLY
1-Motor Control PCB
2-Harness Assy
- ⑦

314127-01 NWT HOUSING SPINDLE ASSEMBLY
1-Housing Base
2-Spindle Assy
3-L/R Guide Shafts
4-Brass Pulley
- 314128-01 NWT DRIVE BELT
- 314129-01 NWT HARDWARE
1-Assorted Screws
2-Zero Stop Tab
3-Tension Spring
- 314174-01 UNIV Replacement Load Pad



NEWTRONICS MOTOR CONTROL SCHEMATIC



SPECIFICATIONS OF THE 1541-II DISK DRIVE

STORAGE

Total formatted capacity	174848 bytes per diskette
Maximum Sequential file size	168656 bytes per diskette
Maximum Relative file size	167132 bytes per diskette
Records per file	65535
Files per diskette	144
Tracks per diskette	35
Sectors per track	17-21
Sectors per diskette	683 total 664 free for data
Bytes per sector	256

INTEGRATED CIRCUIT CHIPS USED

1	6502	microprocessor
	Used for overall control	
2	6522 VIA	Versatile Interface Adapters
	Used for input and output, and as internal timers	
1	6116 RAM	Random Access Memory
	Used as 2K of buffers	
1	16K ROM	Read-Only Memory
	Contains a 16K Disk Operating System (DOS)	

INTERFACE USED

Commodore serial bus with two 6-pin DIN connectors

Device number	selectable from 8-11
Secondary addresses	0-15

PHYSICAL DIMENSIONS

Height	77 mm
Width	184 mm
Depth	256 mm

ELECTRICAL REQUIREMENTS

Three wire-grounded detachable power cable.

Voltage	U.S.	100-120 VAC
	Export	220-240 VAC
Frequency	U.S.	60 HZ
	Export	50 HZ
Power Used		25 Watts

MEDIA

Any good quality 5¼ inch diskette may be used (Commodore diskettes are recommended.)

MAJOR PARTS LIST 1541-II

1	PCB ASSEMBLY 1541-II	340503-01
	CABLE 6P DIN	C1540027-01
	SUB: CABLE 6P DIN	351409-02
	MANUAL — USER	C354139-01
	MANUAL — SERVICE	C314862-01
	DISKETTE DEMO C64	C359800-08
	DISKETTE DEMO 1551/1541	C359800-09
	POWER SUPPLY	C340031-01
	SCREW TAPPING M3X10L (TOP/BOTTOM CASE)	906883-04
	SCREW TAPPING M3X8L (LED/BEZEL) (PCB/SHIELD/BOTTOM CASE)	906883-01
	SCREW TAPPING M3X6L (BRACKET/BOTTOM CASE)	906883-03
	SCREW MACHINE M3X8L (BRACKET/MECHANISM)	906800-03
2	CASE — BOTTOM	C353413-01
3	CASE — TOP	C353412-01
4	BEZEL — NEWT	C353308-01
5	BEZEL — CHINON	C353308-02
6	SHIELD — BOTTOM	351607-01
	CABLE — GROUND	252154-01
7	FOOT — RUBBER	950150-03
8	NAME PLATE	352604-01
9	DRIVE ASSY NEWT D500 (WO/BRACKET)	C359901-01
10	DRIVE ASSY CHINON FZ-501 (WO/BRACKET)	C359902-01
11	DRIVE ASSY TYPE-R (WO/BRACKET)	359901-02
12	BRACKET NEWT	C351523-01
13	BRACKET CHINON	C351523-02
14	KNOB (LEVER) NEWT	C353307-07
15	KNOB (LEVER) CHINON	C353307-06
16	KNOB (LEVER) (N) 02.9 TYPE-R ONLY	C353307-05
17	KNOB (LEVER) (N) 03.0 TYPE-R ONLY	C353307-04
	GEOS DESKTOP SOFTWARE SUB-ASSY	315841-01

PCB-DRIVE ASSEMBLY REFERENCE 1541-II

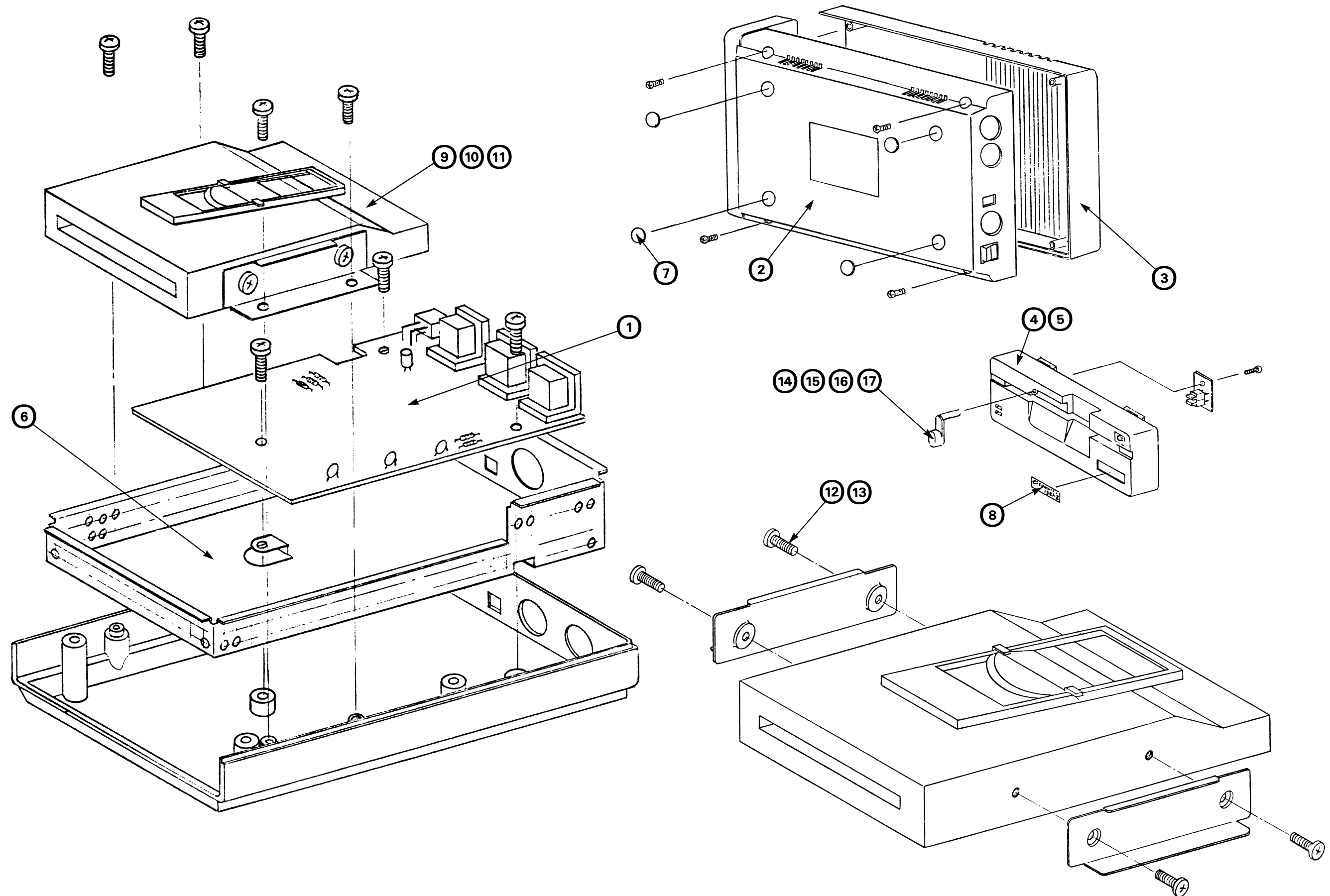
1541-II PCB ASSEMBLY PN# 340503-01 HAS THREE (3) REVISION NUMBERS.

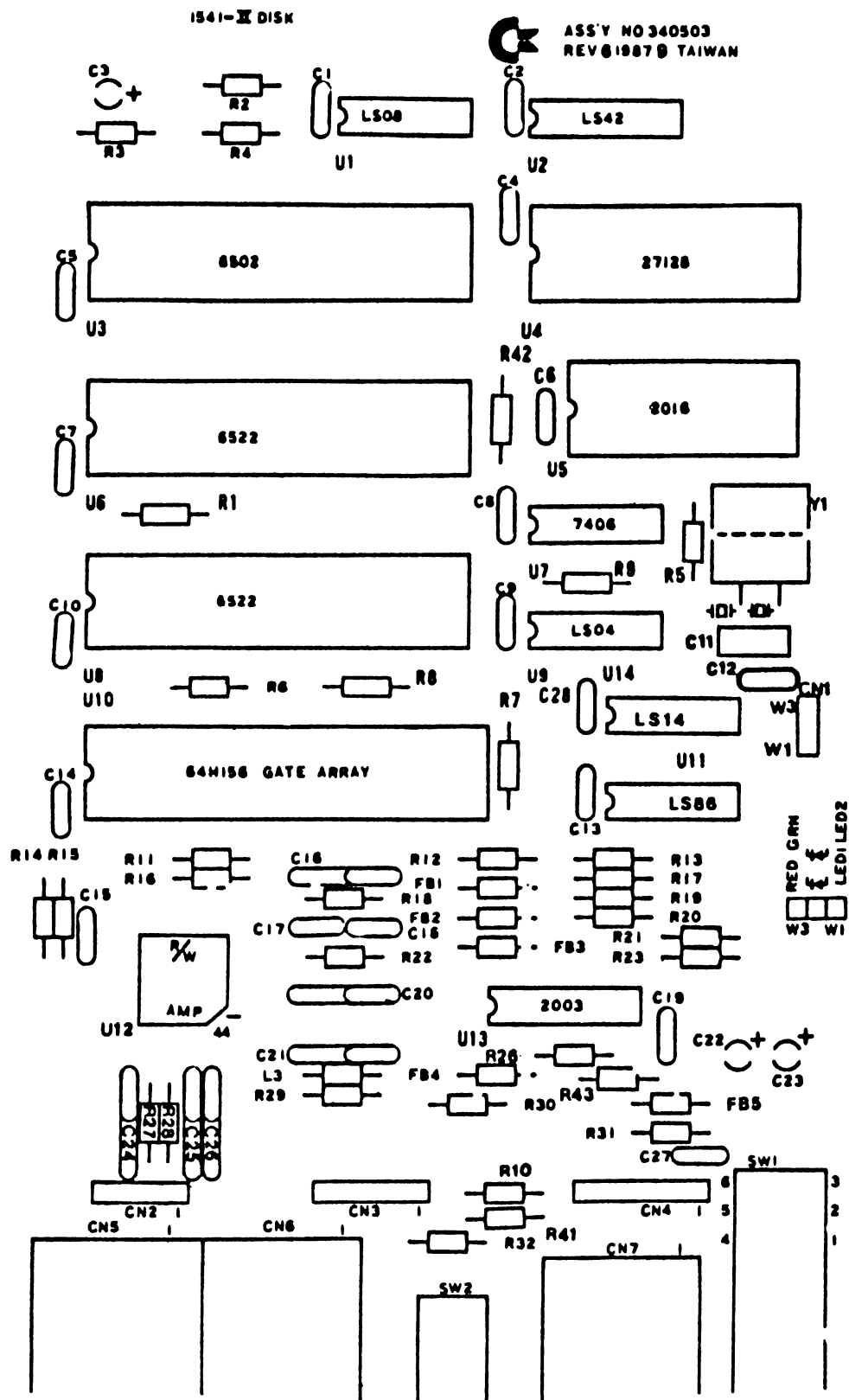
- 1) 340503-01 REV 4 BOARD: CN4 HAS ALL SEVEN (7) PINS.
- 2) 340503-01 REV 5 BOARD: PINS 1 & 2 OF CN4 ARE REMOVED
R41 IS REMOVED.
- 3) 340503-01 REV 6 BOARD: PINS 1 & 2 OF CN4 ARE REMOVED
R41 HAS BEEN RE-INSTALLED.

1541-II HAS THREE (3) DRIVE ASSEMBLIES.

- 1) 359901-01 NEWTRONICS D500 — USE ONLY REV 4 PCB.
- 2) 359902-01 CHINON FZ501 — USE REV 4, 5, OR 6 PCBs.
- 3) 359901-02 NEWTRONICS D500 TYPE R — USE REV 4, 5, OR 6 PCBs.

1541-II SERVICE REFERENCE DIAGRAM



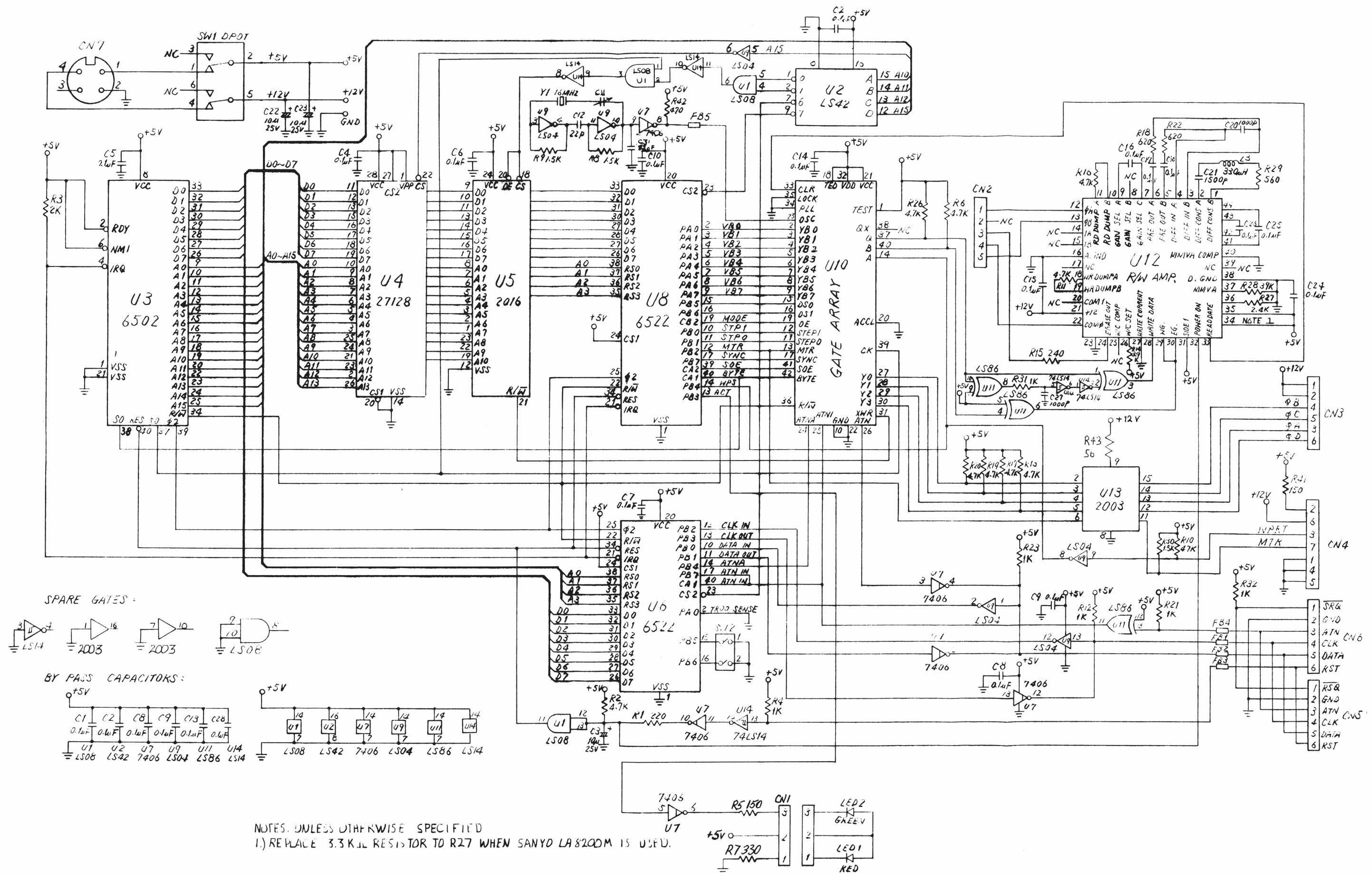


BOARD LAYOUT

COMPONENT PARTS LIST

Commodore part numbers are provided for reference only and do not indicate the availability of parts from Commodore. Industry standard parts (Resistors, Capacitors, Connectors) should be secured locally. Approved cross-references for TTL chips, Transistors, etc. are available in manual form through the Service Department, order #314000-01.

INTEGRATED CIRCUITS			CAPACITORS		
U2	74LS42 DEC	901521-17	C12	RAD CER 50V 10% SL 22P	251072-14
U11	74LS86 EXL/OR	901521-32	C20,27	RAD CER 50V Z5U 1000P	900010-25
U9	74LS04 INV	901521-02	C21	RAD CER 50V 10% Y5P 1500P	251069-10
U1	74LS08	901521-03	C1,2,4-10,13-19,24-26,28	RAD CER 16V Z5U 0.1 U	252036-02
U7	7406	901522-06		Sub: RAD CER 25V Z5U 0.1 U	900010-61
U3	MPU 6502	901435-01		ELEC 25V 10 U	900100-01
U6,8	VIA 6522	901437-01	C3,22,23	TRIMMER 7.5-50pF	251029-04
U4	DOS ROM	C251968-03	C11	Sub: TRIMMER 6.8-45pF	251029-06
U5	S-RAM TMM 2016P	325502-03	MISCELLANEOUS		
U10	64H156 GATE ARRAY	251828-01	Y1	CRYSTAL 16MHZ	900556-02
	Sub: GATE ARRAY	251828-02		Sub: CRYSTAL 16MHZ	900557-01
U13	MPA 2003C	C251871-01	FBI-5	FERRITE BEAD	903025-01
U12	FDD R/W AMP (CX20185)	C252308-01	L1	CHOKE 330 UH	252369-39
	Sub: FDD R/W AMP (MC2871P)	252308-02	SW1	SWITCH ROCKER	252182-01
	Sub: FDD R/W AMP (LA8200M)	252308-03	SW2	SWITCH DIP 2 POS	252144-02
U14	74LS14	901521-30	CONNECTORS		
U4	Socket, 28 Pin	904150-05	CN2	HEADER 4 PIN	251065-04
RESISTORS — All are 5%, ¼W — unless otherwise noted			CN4	HEADER 5 PIN	325562-05
R4,12,21,23,31,32	1K	901550-01	CN3	HEADER 6 PIN	325562-06
R5,41	150 OHM	901550-89	CN4	HEADER 7 PIN	325562-07
	(R41 used W/ FDD 359901-02 only)			Sub: Use W/ FDD 359901-02 only	
R8,9,30	1.5K OHM	901550-69	CN5,6	6 PIN SHIELDED	903361-02
R3	2K OHM	901550-53		Sub: 6 PIN SHIELDED	903361-03
R1	220 OHM	901550-52	CN7	4 PIN SHIELDED	390272-01
R15	240 OHM	901550-29	LED 1	RED 1 x 5mm	355810-01
R37	2.4K OHM	901550-85	LED 2	GREEN 1 x 5mm	355810-02
R7	330 OHM	901550-14			
R14	3.9K OHM	901550-39			
R28	39K OHM	901550-21			
R2,6,11,13,16,17,19,20,26	4.7K OHM	901550-19			
R10	47K OHM	901550-22			
R29	560 OHM	901550-30			
R18,22	620 OHM	901550-40			
R42	470 OHM	901550-58			
R43	56 OHM	901550-131			





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West Chester, PA 19380



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